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The Efficacy of Virtual Reality-Based Intervention for Breast Cancer Patients Symptom and Rehabilitation Management: A Systematic Review and Meta-Analysis

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The Efficacy of Virtual Reality-Based Interventions for Breast Cancer Patients Symptom and Rehabilitation Management: A Systematic Review and Meta-Analysis

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Abstract

Objectives In the care of patients with breast cancer, virtual reality technology is used widely. We sought to assess the effect of Virtual Reality -based intervention on the symptoms and rehabilitation management in patients with breast cancer.

Design Systematic review.

Study selection We included all eligible randomized controlled trials, case-controlled trials, and quasi-experimental studies (published in English and Chinese).

Data sources We searched PubMed, Embase, CENTRAL, and SinoMed four electronic databases, covering the period from database establishment time to November 2020.

Data extraction Two reviewers extracted general information and content data related to the prespecified framework.

Results: A total of 8 studies met the eligibility criteria and were included in the qualitative synthesis. The intervention based on virtual reality has been statistically significant for reducing the anxiety, depression, pain, and fatigue symptoms of patients and improving the shoulder range of motion and cognitive function of breast cancer cancers. Interventions improved anxiety (MD=-8.48, 95%CI= [-13.04 to -3.92], I^2= 84%), depression (MD=-4.27, 95%CI= [-4.63 to -3.92], I^2= 0%), fatigue

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(SMD=-0.92, 95%CI [-4.47 to 2.62], I^2= 99%), pain (SMD=-1.32, 95%CI [-2.56 to -0.09], I^2= 87%), four sections of cognitive function, and the range of motion of shoulder (details attached in this paper).

Conclusions: As a new technology, virtual reality technology has an incomparable effect on managing breast cancer patients' symptoms and rehabilitation. Although there are a series of interventions for breast cancer patients, there are very little researches on interventions using virtual reality technology. Although the evidence so far is limited, the scope of future use is promising. This review challenges the scientific community to design and test interventions based on virtual reality, focusing on the untapped potential for improving the management of breast cancer patients

Keywords virtual reality; breast cancer; meta-analysis

Strengths and limitations of this study

Virtual reality technology is a new technology, and its application to breast cancer patients has broad prospects;

This is the first systematic review aimed at investigating the effectiveness of virtual reality in breast cancer patients to evaluate the relative safety and effectiveness of this technology.

The utility mechanism of virtual reality in applications is not yet clear;

Significant heterogeneity among study results may come from various sources;

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The quality of the included literature may be limited, the long-term effect on patients is not apparent, and there is a lack of literature support with a higher level of evidence.

Systematic review protocol and registration None.

Patient and public involvement No patient involved.

1.Introduction:

Breast cancer is the most commonly occurring cancer in women(1). There were over 2 million new cases and nearly 630 thousand deaths globally in 2018(2). Moreover, according to the prediction, new topics will increase by more than 46% by 2040(3). In recent years, there is a remarkable improvement in breast cancer-related survival rate and life quality(4), because of the accessibility to various options of treatment and rehabilitation, for instance, early breast cancer screening, chemotherapy radiotherapy, surgical therapy, and comprehensive rehabilitation management(5-9). However, the long-term breast cancer disease progression and the treatment have some adverse reactions, and these can cause some physical and psychological negative consequences(10). These possible long-term adverse reactions include somatic symptoms such as limited joint range of motion, pain, and psychological distress such as fatigue, anxiety, and depression(11-13). Simultaneously, these symptoms and negative emotions may further cause the patient's cognitive

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dysfunction, leading to poor predictive outcomes of the patient and affecting all aspects of their life(14, 15).

To improve symptoms, side effects, and the quality of life, some adjuvant treatments are used for signs and rehabilitation management of breast cancer patients(16). This process is often carried out through traditional physical therapy and occupational therapy to improve the patients' symptoms and perform rehabilitation management(17). However, these methods are time-consuming, cumbersome, and require many human resources, and the results often depend on the medical staff's actual capabilities (18). Moreover, traditional rehabilitation therapy is primarily mechanical repetitive training, which lacks interest and motivation(19). Patients often lose interest in this process, which affects the training effect of patients. So the limitations of traditional rehabilitation training prompted the emergence of new and effective methods.

With the rapid development of computer science, more and more modern technologies are applied to managing symptoms and rehabilitation of breast cancer patients, and it is continuously developing(20).

Virtual reality (VR) is mainly used as a new type of rehabilitation therapy for breast cancer patients. The concept of VR was first proposed in the 1960s and gradually rose at the end of the 20th century, and it is an artificial intelligence technology based on multidisciplinary cooperation(21).

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It uses the computer as the core architecture, calculates and simulates a three-dimensional virtual environment, and combines with other somatosensory devices (such as somatosensory handle, sensing helmet, 3D glasses) to make up the gap between the virtual environment and reality(22).

It allows interactive feedback between the user and the virtual environment, providing the user with visual, auditory, tactile, and other signal stimuli to produce their feelings. The features of virtual reality are summarized into '3i'. They are immersive, interactive, and imaginative (23). Compared with the traditional " viewing from the outside - in silico operation, "VR places more emphasis on human beings' dominant role in virtual environments. With the high-speed development of virtual reality technology, its connotation is also further broadened. Based on Virtual Reality (VR), the concepts of Augmented Reality (AR), Mixed Reality (MR) were extended(24). Although virtual reality technology is more diversified, it is mainly applied to breast cancer patients in immersive experiences using headgear and computer games.

In recent years, VR has received increasing attention as a symptom and rehabilitation training approach to address pain reduction, fatigue, anxiety, depression, and cognitive dysfunction in different groups. It does have promising results for physical rehabilitation management of some diseases, such as stroke, Parkinson's, and patients' pain management

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(25-27). As for breast cancer patients, virtual reality technology was gradually introduced into American scholars' clinical intervention research in the early 21st century(28). Existing systematic reviews primarily focused on the inclusion and analysis of mixed samples of cancer patients(29), but breast cancer patients' actual effect has not yet been specifically analyzed.

Therefore, we need to focus on specific populations of patients to be selected in research. Thus, doubts could be raised as to whether VR treatment methods for breast cancer can improve these symptoms; and whether the rehabilitation of patients is effective is worth studying. For this reason, we aimed to perform a meta-analysis as a scientific method to test the effects of uncertain treatment methods using statistical methods to examine whether VR training is practical for breast cancer patient's symptoms and rehabilitation management. Based on the data analysis findings, we can make an objective evaluation and analysis of virtual reality technology in the rehabilitation and symptom management of breast cancer patients, hoping to provide a reference for the Application of patients and the clinical decision-making of medical staff.

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2. Methods:

2.1 Search strategy

We checked up to three English databases (Pubmed, Embase, and CENTRAL), and one Chinese database (SinoMed) from the establishment of databases to November 2020, with the following Mesh terms and text words: ("Breast Neoplasms" OR "Breast Cancer Lymphedema" OR "breast tumor*" OR "breast cancer") AND ("Virtual Reality" OR "Virtual Reality Exposure Therapy" OR "simulation environment"). Appendix A shows the exact searching strategy. Finally, based on the above results, we performed a snowballing approach to search, screen, and examine papers, classifying them as potentially eligible studies.

2.2 Eligibility criteria

2.2.1 Types of studies:

We included all eligible randomized controlled trials, case-controlled trials, and quasi-experimental studies (published in English and Chinese), at least one of which was an ongoing virtual reality training program for inclusion in the review.

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2.2.2 Types of participants:

Our inclusion population was required to meet the following criteria: i. breast cancer diagnosed by pathological analysis (stage 0–IV); ii. age ≥ 18 years old; iii. ongoing or completed cancer treatment. We also developed exclusion criteria for the participants: i. brain primary or metastatic disease (patients may have adverse reactions to VR); ii. combined with other severe diseases (such as severe heart disease, hypertension.); iii. previous psychiatric illness or cognitive impairment; iv. primary limb movement disorders; v. visual and auditory impairment.

2.2.3 Types of interventions:

The trial group participated in the organized and systematic intervention project based on virtual reality technology for breast cancer patients. The primary intervention goal was to reduce the psychological disorder, dysfunction, and pain of patients.

The intervention experiment based on virtual reality needs to meet the following core conditions:

1.Using computer-generated human-computer interactive intervention means in the virtual environment, such as VR glasses and headwear equipment;

2. Patients immerse themselves in the virtual environment constructed by electronic equipment;

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3. Patients can perceive, feel, and interact in a manner that is similar to a physical place;

4. They can achieve effects by combining stimulation over multiple sensory channels such as the vision, sound, touch, and perception produce a corresponding stimulus.

2.2.4 Types of outcomes:

Due to the different focus of the original research, the leading evaluation indicators may be various. According to the purpose of this study, we selected the outcome indicators of this study. The results included: the patient's anxiety, fatigue, depression, pain, cognitive function, and motor range.

2.3 Data collection and analysis

2.3.1 Selection of studies

Two searchers (HYZ, QSZ) independently screened all search results (title, abstract) in the database to identify studies that could be included in this paper. After initial screening, HYZ and QSZ performed a full-text evaluation to determine the literature for inclusion. When disagreements occurred, HYZ and QSZ discussed the disparities and resolved them. If consensus was not reached after discussion between the two, it was decided by HX's independent arbitration. If necessary, we contacted study

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authors for specific information.

2.3.2 Data extraction and management

Two review authors (HYZ, QSZ) independently extracted data onto a predesigned data collection form, including the trial setting, inclusion and exclusion criteria, study population, interventions, outcome measures, and final results of the papers. Both review authors, before data extraction, were given detailed instructions and participated in training sessions. When disagreements appeared, they were resolved by discussion among the two review authors (HYZ, QSZ) or HX's independent arbitration. We contacted study authors for clarification if necessary.

2.2.3 Assessment of risk of bias in included studies

The quality of the studies included in this meta-analysis was evaluated using the quality critical assessment list of randomized controlled trials recommended by the Cochrane Handbook. The index consists of six items on random allocation, allocation concealment, blinding, data completeness, selection reporting bias, and baseline comparability. The items on the list were rated as "met," "unmet," or "unclear." Because the participants of this intervention are all patients, it is theoretically difficult to be double-blinding. Therefore, in our review, an appropriate

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single-blind outcome evaluator was considered to meet the criteria. According to the Cochrane Handbook, we have scoured the literature. It was scored as 1 point if it met the criteria, and the others are not scored. The total scores were 6 points, and more than 3 points could be included in this paper. The evaluation process was conducted by two independent reviewers (HYZ and QSZ). The inter-rater agreement between two raters would be calculated using Cohen's Kappa. If there were an analysis, the two examiners would negotiate and reach a consensus. If there were still differences after negotiation, the third-party examiner (HX) made the final decision.

2.3.4 Measures of treatment effect

We performed data description analysis based on selected outcome measures such as pain, anxiety, depression, joint range of motion, quality of life, cognitive function, number of cases of lymphedema. For continuous variables, the data in the literature we extracted were all mean difference (MD) and standard deviation (SD), so we did not make another conversion. Nevertheless, in some of them, the size of the values resulted from different evaluation tools. We calculated the standardized mean difference (SMD) as needed for the normalization of the data. All trial groups performed a 95% confidence interval (95% CI) and weight in parallel. Cochrane's Review Manager 5.4 (Review Manager 2020)

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software for our analyses for all of the above statistical analyses.

2.3.5 Dealing with missing data

There were no missing data of all these included articles. We did not contact the authors to retrieve their unpublished study data.

2.3.6 Assessment of heterogeneity

We used the chi-square test for all data before conducting heterogeneity analysis across studies. If $I^2 < 50\%$, we considered individual studies' homogeneity, selected the fixed-effect model for statistical analysis.

If $I^2 \geq 50\%$, we considered heterogeneity among studies, but according to clinical judgment, there was consistency among studies, and effect size needed to be pooled, so we selected a random effect model. Forest plot graphics were generated to present the pooled effect. All tests are two-sided, and P-value<0.05 was considered to be statistically significant.

3. Results:

3.1 Search process

After searching in 4 databases, we retrieved 3,058 articles. For these documents, we conducted a duplicate check, preliminary screening, and full-text reading for screening, and finally, eight papers met the criteria of

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this meta-analysis. Figure 1 shows the detailed search process and results.

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Table 1. Risk of bias assessment and quality of the included studies.

Study	Selection bias		Blinding	Attrition bias	Reporting bias	Other sources of bias	Overall
	Randomization	Allocation concealment		Incomplete outcome data	Selective reporting	Baseline comparability	
House et al(30) (USA/2016)	?	?	+	+	+	-	3
Jimenez et al(31) (Australia/2018)	?	?	?	+	+	+	3
Mohammad et al(32) (Jordan/2018)	+	+	+	+	+	+	6
Chirico et al(33)	?	?	-	+	+	+	3

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(Italy/2019)							
Feyzioğlu et al(34) (Turkey/2020)	+	+	+	+	+	+	6
Jin et al(35) (China/2018)	+	?	-	+	+	+	5
Jin et al(36) (China/2018)	-	?	?	+	+	+	3
Chen et al(37) (China/2018)	?	?	-	+	+	+	3

‘‘+’’ = criteria ‘‘met’’; ‘‘-’’ = criteria ‘‘unmet’’; ‘‘?’’ = criteria ‘‘unclear’’. (If it met the criteria, the score is 1 point, otherwise, no score is given.)

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Table 2. Characteristics of the included studies

Study (Country/year)	Design	Participants	Intervention Duration	VR Method	Outcome Measurement	Main Results
House et al(30) (USA/2016)	Pre-post test	Community-dwelling women (n = 6) with post-surgical breast cancer pain in the upper arm.	VR-based treatment regimens are 20-50 min long per session and 2 times per week for a total of 8 weeks.	The BrightArm Duo Rehabilitation System consists of a low-friction robotic rehabilitation table. A computerized forearm supports a display, a laptop computer for the therapist station, a remote clinical server, and a library of custom integrative rehabilitation games.	BDI-II; BVMT-R; TMT; NAB; NRS; Mechanical Goniometers; HVLT-R.	Pain intensity showed significant decreased ($P < 0.05$). Symptoms of depression decreased. Cognitive function improved post-training. Internal rot ROM of the affected shoulder increased ($P<0.01$).
Mohammad	RCT	80 female	Give VR immersive	The CG (n=40) did not receive VR	VAS; SAI.	Compared with the CG,

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et al(32) (Jordan/2018)		patients with breast cancer at a specialized cancer center in Jordan	intervention while receiving morphine treatment.	treatment. The VG(n=40) chose from two scenarios, which included deep-sea diving "Ocean Rift," or sitting on the beach with the "Happy Place" track.		the pain and anxiety of the patients in VG were significantly reduced(all P<0.05).
Chirico et al(38) (Italy/2019)	RCT	94 breast cancer patients were receiving chemotherapy at the National Cancer Institute-IRCCS "Fonda-zone G.	Patients use the VR device during each chemotherapy. The chemotherapy duration is 45-90 minutes, and the device should be used for more than 20 minutes.	The CG(n=34) received routine care, and the MTG(n=30) received music therapy based on CG. The VG(n=30): The VR equipment consisted of a headset and ahead movement tracking system. Participants explore the island through the relaxing landscape created on the Second	SAI; SV-POMS.	Compared with the MTG and the CG, the VRG can significantly improve the patients' anxiety, depression, and fatigue.

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		Pascale," Naples, Italy.		Life® platform through forests, animal observation, mountain climbing, and swimming.		
Feyzioğlu et al(34) (Turkey/2020)	RCT	40 women with breast cancer had undergone unilateral mastectomy with axillary lymph node dissection and were receiving adjuvant therapy.	Both groups received treatment for 45 minutes each time, 2 times a week, for 6 weeks. Outside the treatment period, all patients received the same home exercise program.	The VG(n=20) received tissue massage, passive mobilization, and Xbox360Kinect™ video game treatment. The patient performed activities based on VR boxing, darts, and other games; the CG(n=20) received standard upper limb physical therapy treatment, including scar tissue massage and passive mobilization.	VAS; Digital Goniometer.	The two groups of patients' pain and ROM scores changed significantly after treatment ($p<0.01$).

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Jimenez et al(31) (Australia/2018)	RCT	37 breast cancer patients receiving radiotherapy at a targeted cancer treatment center.	The VG accepted one-hour health education based on virtual reality technology each time, a total of 18 times.	The VG(n=19) used the VERT system (Version 2.9) to provide basic technology, anatomical knowledge, and radiation dose information about RT for patients. The CG(n=18) received regular health education on radiotherapy knowledge.	STAI.	For both groups, anxiety levels were decreased at time points before simulation, at the beginning of treatment, and at the end of treatment ($p > 0.05$).
Jin et al(35) (China/2018)	RCT	120 cases of breast cancer patients undergoing surgery in the hospital.	All patients started to exercise their affected limbs 6 hours after the operation and were trained 6 times a week for 30 minutes a day for a total of 3 months.	The VG(n=60) used a professional rehabilitation platform; through the virtual and VR somatosensory equipment (helmet, joystick, data gloves), patients can choose their rehabilitation games for training. The CG(n=60) received traditional	SF-36	The scores and average scores of all dimensions of SF-36 in the VG were higher than those in the CG ($P<0.05$).

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				rehabilitation training.		
Jin et al(36) (China/2018)	RCT	76 patients received modified radical mastectomy for unilateral or bilateral breast cancer in the hospital.	Both groups of patients received three-month rehabilitation training twice a day, 15-30 minutes each time.	The VG(n=38) selected appropriate training methods for rehabilitation according to the patients' muscle strength characteristics in different postoperative rehabilitation periods. The system includes sensors (HTC), upper limb rehabilitation devices, and a game platform (Unity 3D). The CG (n=38) received routine rehabilitation training under the guidance of nurses.	Mechanical Goniometers; No elastic measuring tape.	The range of motion of the shoulder joint in the VG was significantly better than that of the CG (P<0.01); the rehabilitation training of the VG was better than the CG in the degree of edema of the affected limbs(P<0.05).
Chen et al(37) (China/2019	Pre-post test	80 breast cancer patients hospitalized for	All patients were trained once a day and 5 times a week for 8	The patient used a VR-based cognitive rehabilitation system, allowing the patient to promote the	MoCA	The patient's cognitive function has been significantly improved

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)		chemotherapy.	weeks.	patient's cognitive function rehabilitation through language training, memory training, and performing specific tasks in a virtual environment.		(P<0.05).

Abbreviations: BDI-II: Beck Depression Inventory, Second Edition; BVMT-R: Brief Visuospatial Memory Test, Revised; CG: Control Group; HVLT-R: Hopkins Verbal Learning Test, Revised; MoCA: Montreal Cognitive Assessment; MTG: Music Therapy Group; NAB: Neuropsychological Assessment Battery; NRS: Numeric Pain Rating Scale; RCT: Randomized Controlled Trial; ROM: Range of Motion; SAI: State Anxiety Inventory; SF-36: 36-Item Short Form Survey; STAI: State-Trait Anxiety Inventory; SV-POMS: Short Version of Profile of Mood States; TMT: Trail Making Test; VG: Virtual Reality Group; VAS: Visual Analog Scale.

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3.2 Critical appraisal of quality

This meta-analysis used the quality and bias risk assessment tools recommended by the Cochrane Handbook(39). On this basis, we made improvements, using a scoring method to evaluate the risk of bias in the literature to reflect the quality of the literature better. Table 1 shows the analysis results of each study. Three studies had a low risk of bias, and the other five studies had a high risk of bias. The main reasons were as follows: randomization, allocation concealment method, blinding was not clear or wrong.

3.3 Study Characteristics

The essential characteristics of the included studies are shown in Table 2. The quality of the eight studies met our inclusion criteria. Three studies were from China (35-37), and the others were from the United States(30), Jordan(32), Italy(33), Turkey(34), and Australia(31). 2 of them were before and after controlled trials, and the others were randomized controlled trials. These researches were conducted in the specialized ward and cancer center of the hospital. The number of participants ranged from 6 to 120. All participants were breast cancer patients who met the inclusion criteria. VR interventions included both immersive and non-immersive formats, and the duration of the intervention ranged from

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20 minutes to 3 months. These studies assessed VR-based interventions' impact on health-related outcomes, including anxiety, depression, fatigue, pain, cognitive function, quality of life, and upper limb joint motion range.

3.4 Outcome analysis

3.4.1 Anxiety meta-analysis

Three studies evaluated the impact of VR on breast cancer patients' anxiety. Compared with the control group, the results showed that VR-based intervention has a significant effect (MD=-8.48, 95%CI= [-13.04 to -3.92], P=0.0003, Figure 2). Heterogeneity was considerably high (I^2= 84%).

3.4.2 Depression meta-analysis

Two studies explored the impact of VR intervention on the depression of breast cancer patients, and the results showed that the VR group had a better effect than the control group, the pooled effect size was statistically significant (MD=-4.27, 95%CI= [-4.63 to -3.92], P<0.00001, Figure 2). No obvious heterogeneity between studies (P=0.84, I^2= 0%).

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3.4.3 Fatigue meta-analysis

Two studies evaluated the impact of VR on the fatigue of breast cancer patients. Due to different evaluation tools, SMD was selected for the combination of effect sizes. There was apparent heterogeneity between the studies (I^2= 99%). A random-effect model was chosen for analysis. The results showed that VR could not improve the patient's fatigue, the combined effect size was not statistically significant (SMD=-0.92, 95%CI [-4.47 to 2.62], P=0.61, Figure 2).

3.4.4 Pain meta-analysis

Three studies evaluated the impact of VR on pain of breast cancer patients. Due to the different evaluation tools, SMD was selected for the combination of effect sizes. There was apparent heterogeneity among the studies (I^2= 87%). A random-effect model was chosen for analysis. The results showed that VR could significantly improve the patient's pain degree, and the pooled effect size was statistically significant (SMD=-1.32, 95%CI [-2.56 to -0.09], P=0.04, Figure 2).

3.4.5 Cognitive function meta-analysis

Two studies evaluated the impact of VR on breast cancer patients' cognitive function from four aspects: memory, visual function, language function, and executive function (Figure 3). In terms of memory, the results showed that the pooled effect size was statistically different

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(SMD=2.32, 95%CI [1.93 to 2.72], P<0.00001), and there was no significant heterogeneity among the studies (P=0.33, I^2= 0%). While in terms of visual function (SMD=2.41, 95%CI [1.41 to 3.42], P<0.00001), language function (SMD=0.87, 95%CI [-0.44 to 2.19], P=0.19), and executive function(SMD=1.64, 95%CI [-0.69 to 3.96], P=0.17), the results showed that the pooled effect size was statistically different in visual function and there were no statistically different in language function, executive function, the heterogeneity of each study was existed(I^2= 55%; I^2= 80%; I^2= 93%).

3.4.6 Range of motion of upper limb joint meta-analysis

Three studies evaluated the effect of VR on the range of motion of breast cancer patients' shoulder joints (Figure 4). They were evaluated from flexion, extension, abduction, adduction, internal rotation, and external rotation. In terms of flexion and extension, the results showed that the pooled effect size showed that VR and the traditional rehabilitation had the same effect of improving the patient's joint range of motion (Flexion: MD=23.75, 95%CI [-6.03 to 53.53], P=0.12; Extension: MD=16.86, 95%CI [-27.18 to 60.90], P<0.00001), and their heterogeneity were greater (Flexion I^2=96%; Extension I^2=93%). In terms of adduction, abduction, internal rotation, and external rotation, the results showed that the combined effect size showed that VR has a significant effect on improving the patient's joint mobility (Adduction: MD=15.54, 95%CI [12.79 to18.29], P<0.00001; Abduction: MD=17.49, 95%CI [13.03 to

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21.96], P<0.00001; Internal rotation: MD=7.09, 95%CI [4.85 to 9.32], P<0.00001; External rotation: MD=1.07, 95%CI [0.69 to 1.45], P<0.00001), and their heterogeneity were small or no obvious heterogeneity(Adduction I^2=0%; Abduction I^2=0%; Internal rotation I^2=0%; External rotation I^2=31%).

4.Discussion

This systematic review evaluated the application effect of intervention measures based on virtual reality in breast cancer patients. Consistent with previous research (38), the results showed that compared with traditional breast cancer rehabilitation management, patients' anxiety, depression, fatigue, pain, cognitive function, and shoulder range of motion were improved in varying degrees. In some patients' anxiety, fatigue, pain, and cognitive function, the pooled effect size showed that VR intervention improved these symptoms. Still, at the same time, the statistical test results showed that there was significant heterogeneity between these data. The combined impact showed that VR intervention was ineffective in the range of motion of shoulder joint flexion and extension. Still, in the other four joints, it had significant statistical significance. Although the data results showed no significant statistical difference between VR-based intervention methods, this did not mean that this intervention method was ineffective. This may be due to the small sample size of the included study.

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In these studies, VR-based intervention used the immersive experience to intervene and achieved positive results. These devices specifically included: wearable helmets, 3D glasses, somatosensory operating devices, smart displays, and computers. After these devices are connected, they can interact with the outside world in real-time. Subjects can interact cognitively or physically in a computer-generated virtual environment to achieve their expected results. Therefore, compared with the traditional therapy in the control group, it can achieve better results. The primary characteristic of VR is immersion, which can enhance the subject's heart rate and activate brain circuit functions. In particular, it acts on the midline θ of the brain's forehead, including the frontal lobe and the left temporal area, bringing a pleasant emotional experience to subjects, activating the attention of related brain networks, and improving motor integration and spatial orientation functions(40). An immersive tool can change patients' awareness in a virtual environment, provide a positive and pleasant experience, improve patients' cognitive function, and improve breast cancer patients' intervention effect. Because of the interactivity of VR, patients can enhance the effectiveness of an intervention in the real-time interaction of VR. This may be related to the activation of the midbrain marginal dopaminergic pathway and VR technology stimulation in the interaction between patients and VR(41). The brain's reward mechanism can improve the attractiveness of participating in the intervention, enhance breast cancer patients' interest, and make patients more like this form. VR has real-time solid subjects in

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the virtual environment. Not only can real-time monitor the behavior and physiological response of patients, background data operation and guardians observe the transmission of stimulation in the virtual environment and comprehensively control the duration and intensity of rehabilitation(42). Simultaneously, it creates a relatively safe range of activities, allowing patients to carry out rehabilitation tasks in a relatively stable environment. The above utility mechanism improves the positive experience, attention, interaction, and participation of patients to a certain extent. It carries out rehabilitation in a real-time dynamic monitoring and relatively safe environment to improve patients' compliance, promote the rehabilitation of patients with breast cancer, and improve their prognosis.

The above research discussion shows that VR-based breast cancer symptom management and rehabilitation intervention methods are worth learning and promoting. With the rapid development of 5G communication technology(43), artificial intelligence technology, and Internet of Things technology, VR-based breast cancer intervention methods are bound to be more widely used and will receive technological innovation. Around the world, some countries have begun to formulate relevant laws and policies to give VR technical and economic support and policy inclination(44) so that it can accelerate industrial integration and put the new VR technology into clinical practice in medical and health care as soon as possible.

As an emerging technology, VR has the advantages of high simulation, high immersion, and strong maneuverability. However, since its

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application is still in its infancy, to ensure subsequent promotion and Application, the following issues should be paid attention to: ① The use of VR Security Question. When using a VR device, subjects are immersed in a virtual environment and often ignore the potential dangers in the actual scene. There is a risk of accidental injury and falling. Therefore, they should be monitored during use. At the same time, professionals should evaluate and modify their environment to ensure patient safety. Simultaneously, the articles retrieved in our systematic review rarely involve safety evaluation indicators, so we did not use them as outcome indicators. In the follow-up research, the safety evaluation system of VR application in patients can be constructed and applied. ② Be alert to the potential adverse effects of VR. The highly immersive nature of VR stimulates sensory nerves, which may lead to adverse reactions such as blurred vision, dizziness, nausea, and vomiting. Although the existing short-term intervention studies have not clearly defined it as an adverse event, long-term effect monitoring is still needed to avoid possible health problems and addiction caused by long-term use. ③ The hardware and ecological construction are not yet mature. The Application of VR to breast cancer is still in its infancy. Most VR products on the market are not specifically developed for breast cancer patients. The clinical integration is low, the construction of ecological resources is not yet mature, and patients' targeted demands cannot be met. It can be low selectivity. It is suggested that a multidisciplinary team should be formed to develop VR hardware and ecological communities

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for breast cancer patients, build a more extensive resource library, enrich the functions of VR, and give patients more choices to maximize the benefits. ④During the new COVID-19 pandemic, the social activities of the population, are more alienated. Because of the convenience and mobility of breast cancer intervention based on VR, patients can also be treated at home, reducing the risk of infection when going out and alleviating the feeling of social isolation.

5. Conclusion

The results of this systematic review show that VR-based intervention can effectively improve the anxiety, depression, fatigue, pain, cognitive function, and range of joint shoulder movement of breast cancer patients. In the future, researchers should conduct large-scale pilot studies on the symptom management and rehabilitation of breast cancer patients to explore its applicability and feasibility in breast cancer patients. Medical staff should assess the patient's overall situation and strictly monitor the patient's condition during the Application of VR to ensure patient safety.

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Author Contributions: HYZ is responsible for the collation of literature data and the writing of the full text. At the same time, HYZ and QSZ are responsible for database

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retrieval, literature quality evaluation, and technical route support. HX and ZSZ carried out the final quality evaluation and revision of the paper.

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Competing interests: None declared.

Patient consent for publication: Not required.

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Figure legend/caption: Figure1 Flow chart of selection process; Figure2 Effect of VR-based interventions on anxiety, depression, fatigue and pain; Figure3 Effect of VR-based interventions on cognitive function; Figure 4 Effect of VR-based interventions on ROM of shoulder

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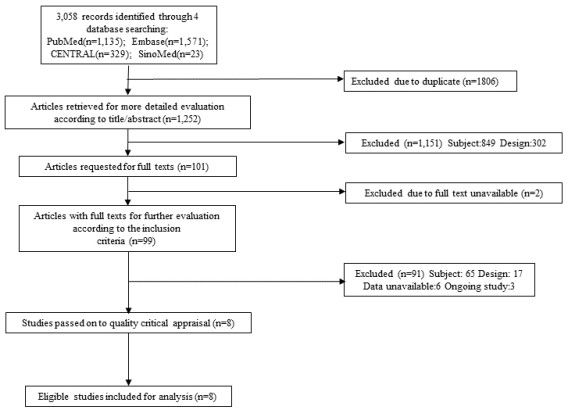


Figure. 1. Flow chart of selection process

Figure1 Flow chart of the selection process

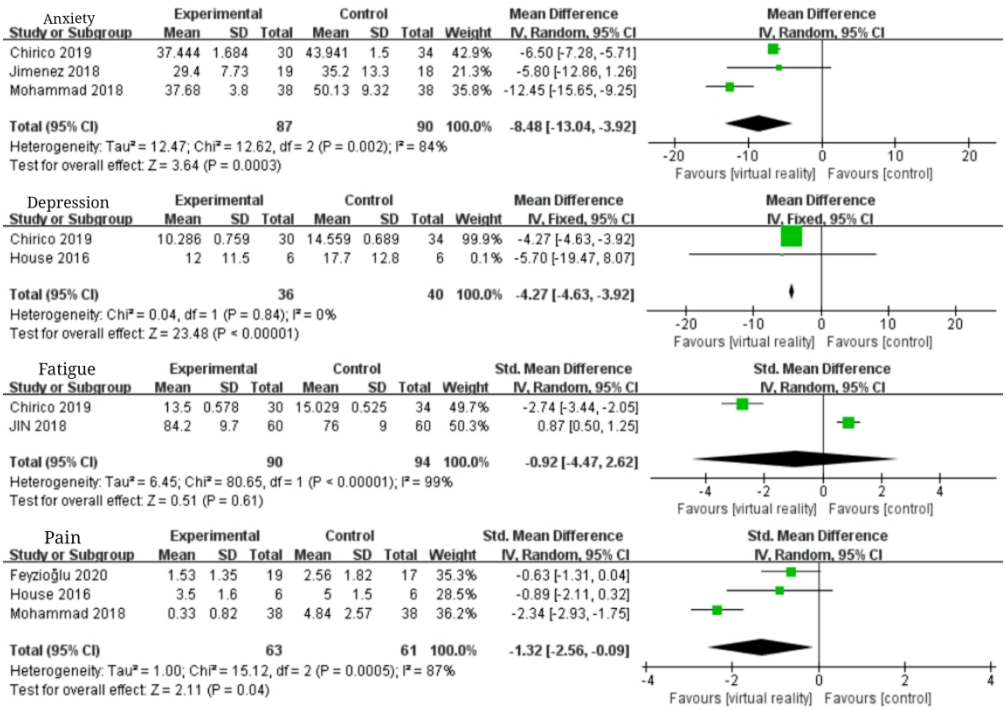


Figure2 Effect of VR-based interventions on anxiety, depression, fatigue and pain

1151x817mm (72 x 72 DPI)

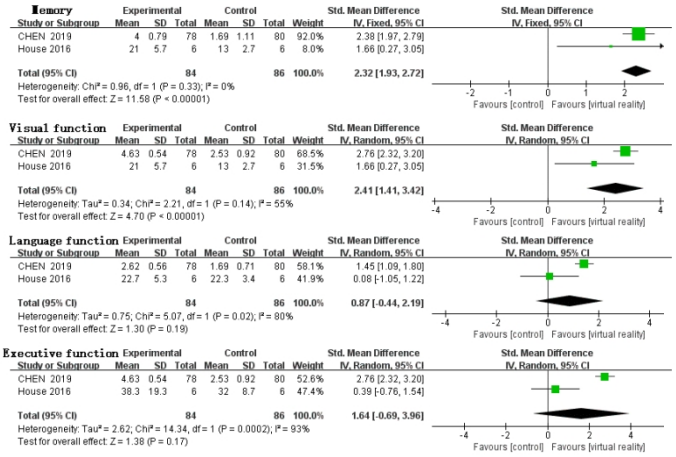


Figure3 Effect of VR-based interventions on cognitive function

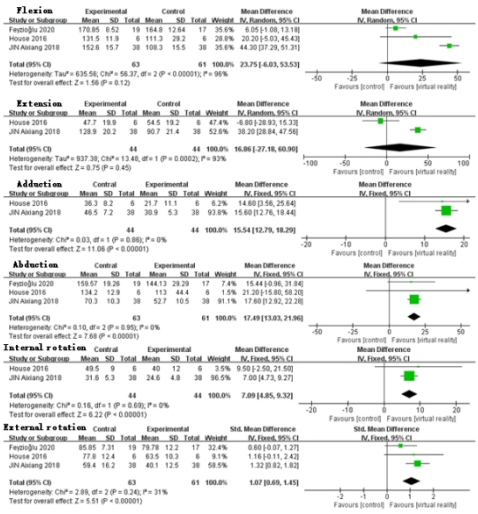


Figure 4 Effect of VR-based interventions on ROM of shoulder

Search Strategy

Appendix 1. PubMed (English)

1. Breast Neoplasms [MeSH Terms]
2. Breast Cancer Lymphedema [MeSH Terms]
3. breast tumor* [Title/Abstract]
4. breast cancer [Title/Abstract]
5. breast carcinoma* [Title/Abstract]
6. mammary gland carcinoma* [Title/Abstract]
7. Virtual Reality [MeSH Terms]
8. Virtual Reality Exposure Therapy [MeSH Terms]
9. User-Computer Interface [MeSH Terms]
10. virtual rehabilitation system [Title/Abstract]
11. virtual reality head-mounted display [Title/Abstract]
12. VR [Title/Abstract]
13. game [Title/Abstract]
14. gaming [Title/Abstract]
15. simulation system [Title/Abstract]
16. simulation environment [Title/Abstract]
17. 1 OR 2 OR 3 OR 4 OR 5 OR 6
18. 7 OR 8 OR 9 OR 10 OR 11 OR 12 OR 13 OR 14 OR 15 OR 16
19. 17 AND 18

Appendix 2. Embase (English)

1. 'breast cancer'/exp
2. 'breast tumor'/exp
3. 'breast cancer-related lymphedema'/exp
4. 'breast neoplasm*' [ab,ti]
5. 'breast carcinoma*' [ab,ti]
6. 'mammary gland carcinoma*' [ab,ti]
7. 'virtual reality'/exp
8. 'virtual reality exposure therapy'/exp
9. 'computer interface'/exp
10. 'virtual reality head mounted display'/exp
11. 'virtual rehabilitation system'/exp
12. 'vr' [ab,ti]
13. 'user-computer interface' [ab,ti]
14. 'game' [ab,ti]
15. 'gaming' [ab,ti]
16. 'simulation system' [ab,ti]
17. 'simulation environment' [ab,ti]
18. 'interactive' [ab,ti]
19. 'reality system' [ab,ti]
20. 1 OR 2 OR 3 OR 4 OR 5 OR 6

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21. 7 OR 8 OR 9 OR 10 OR 11 OR 12 OR 13 OR 14 OR 15 OR 16 OR 17 OR 18 OR 19
22. 21 AND 22

Appendix 3. CENTRAL (English)

1. MeSH descriptor [[Breast Neoplasms] explode all trees
2. MeSH descriptor [[Breast Cancer Lymphedema] explode all trees
3. breast tumor* [ti,ab,kw]
4. breast cancer [ti,ab,kw]
5. breast carcinoma* [ti,ab,kw]
6. mammary gland carcinoma* [ti,ab,kw]
7. MeSH descriptor [Virtual Reality] explode all trees
8. MeSH descriptor [Virtual Reality Exposure Therapy] explode all trees
9. MeSH descriptor [User-Computer Interface] explode all trees
10. virtual rehabilitation system [ti,ab,kw]
11. virtual reality head-mounted display [ti,ab,kw]
12. VR [ti,ab,kw]
13. game [ti,ab,kw]
14. gaming [ti,ab,kw]
15. simulation system [ti,ab,kw]
16. simulation environment [ti,ab,kw]
17. Interactive [ti,ab,kw]
18. reality system [ti,ab,kw]
19. 1 OR 2 OR 3 OR 4 OR 5 OR 6
20. 7 OR 8 OR 9 OR 10 OR 11 OR 12 OR 13 OR 14 OR 15 OR 16 OR 17 OR 18
21. 19 AND 20

Appendix 4. SinoMed (Chinese)

1. Breast Cancer
2. Virtual Reality
3. 1 AND 2

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BMJ Open

The Efficacy of Virtual Reality-Based Interventions for Breast Cancer Patients Symptom and Rehabilitation Management: A Systematic Review and Meta-Analysis

Journal:	*BMJ Open*
Manuscript ID	bmjopen-2021-051808.R1
Article Type:	Original research
Date Submitted by the Author:	02-Dec-2021
Complete List of Authors:	Zhang, Huayi; Zhengzhou University, School of Nursing and Health Xu, Hui; Zhengzhou University, Department of Fundmental Nursing Zhang, Zhen-xiang; Zhengzhou University, Department of Fundmental Nursing Zhang, Qiushi; Zhengzhou University, Department of Fundmental Nursing
Primary Subject Heading:	Oncology
Secondary Subject Heading:	Complementary medicine, Rehabilitation medicine
Keywords:	Breast tumours < ONCOLOGY, Information technology < BIOTECHNOLOGY & BIOINFORMATICS, REHABILITATION MEDICINE

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The Efficacy of Virtual Reality-Based Interventions for Breast Cancer Patients Symptom and Rehabilitation Management: A Systematic Review and Meta-Analysis

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Word count: 4,039 (excluding title page, abstract, references, figures, and tables)

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Abstract

Objectives To determine the effectiveness of VR-based intervention on the symptoms and rehabilitation management in patients with breast cancer.

Design Systematic review and meta-analysis.

Study selection We included all eligible randomized controlled trials (RCTs) and quasi-experimental studies (published in English and Chinese).

Participants Breast cancer patients (≥18 years) undergoing cancer treatment.

Interventions Any intervention administered to improve the symptoms and rehabilitation of breast cancer patients. The control group was given conventional care.

Outcomes All outcomes were as follows: pain, fatigue, anxiety, depressive symptoms, cognitive function, and range of motion of upper limb in breast cancer patients.

Data sources We searched PubMed, Embase, CENTRAL, and SinoMed, four electronic databases, covering the database establishment period to November 2020.

Review methods Two reviewers independently extracted content and data consistent with the prespecified framework and assessed risk bias. Random-effects meta-analysis was used to pool data across trials. Meta-analysis was performed using Review Manager V.5.4.

Results: A total of 8 studies met the eligibility criteria and were included in this study. The combined effect size showed that VR was positive for improving patients' anxiety(SMD=-2.07, 95%CI= [-3.81 to -0.34], I^2= 95%) and abduction of upper limbs(MD=15.54, 95%CI= [12.79 to 18.29], I^2= 0%), but fatigue(SMD=-0.92, 95%CI= [-4.47 to 2.62], I^2= 99%) was not. Qualitative analysis showed VR improved patients' depressive symptoms, pain and cognitive function.

Conclusions: Virtual reality technology has a good effect on breast cancer patients' symptoms and rehabilitation management, but the quality of evidence is low, and the sample size is small. There are few intervention studies completed so far, so we could not make an exact recommendation and conclusion. We have a favorable view of this, and more clinical studies are needed in the future to improve the credibility of the results.

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Keywords virtual reality; breast cancer; meta-analysis

Strengths and limitations of this study

1. Unlike previous systematic reviews that focused on many types of cancer patients, this is the first review that aims at investigating the effectiveness of virtual reality in breast cancer patients to evaluate the relative safety and effectiveness of this new technology.
2. Due to the diversity of research design, we used the random effect model for analysis.
3. Faced with the COVID-19 pandemic period, the intervention based on virtual reality provides patients to undergo rehabilitation effectively at home and reduce the risk of infection.
4. High heterogeneity in some outcome measures due to different measurement tools, low quality of included studies, and small sample size limit the further promotion of the results, resulting in a risk of bias.
5. The quality of the included literature is low, the long-term effect on patients is still unclear, there is a lack of literature support with a higher level of evidence.

Systematic review protocol and registration None.

Patient and public involvement Not applicable.

Ethics Statement Not applicable.

1. Introduction:

Breast cancer is the most commonly occurring cancer in women[1]. There were over 2 million new cases and nearly 630 thousand deaths globally in 2018[2]. Moreover, according to the prediction, new breast cancer patients will increase by more than 46% by 2040[3]. In recent years, there has been a remarkable improvement in breast cancer-related survival rate and life quality[4], because of the accessibility to various treatment options and rehabilitation options, for instance, early breast cancer screening, chemotherapy radiotherapy, surgical therapy, and comprehensive treatment

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rehabilitation management[5-9]. However, the long-term breast cancer disease progression and the treatment have some adverse reactions, which can cause some physical and psychological negative consequences[10]. These possible long-term adverse reactions include somatic symptoms such as limited joint range of motion, pain, and psychological distress such as fatigue, anxiety, and depressive symptoms[11-13]. Simultaneously, these symptoms and negative emotions may further cause the patient's cognitive dysfunction, leading to poor predictive outcomes and affecting all aspects of their life[14,15].

Some adjuvant treatments are used for rehabilitation management of breast cancer patients to improve symptoms, side effects, and the quality of life[16]. This process is often carried out through traditional physical therapy and occupational therapy to improve the patients' symptoms and perform rehabilitation management[17]. However, these traditional methods are time-consuming, cumbersome, and cost a lot, and the results often depend on the medical staff's actual capabilities[18]. Moreover, traditional rehabilitation therapy is primarily mechanical repetitive training, which lacks interest and motivation[19]. Patients often lose interest in this process, which affects the training effect of patients. So the limitations of traditional rehabilitation training prompted the emergence of new and effective methods.

With the rapid development of computer science, more and more modern technologies are applied to managing symptoms and rehabilitation of breast cancer patients, and it is continuously developing[20].

Virtual reality (VR) is mainly used as a brand new rehabilitation therapy for breast cancer patients. The concept of VR was first proposed in the 1960s and gradually rose at the end of the 20th century, and it is an artificial intelligence technology based on multidisciplinary cooperation[21].

It uses the computer as the core architecture, calculates and simulates a three-dimensional virtual environment, and combines with other somatosensory devices (such as somatosensory handle, sensing helmet, 3D glasses) to make up the gap between the virtual environment and reality[22].

It allows interactive feedback between the user and the virtual environment, providing

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the user with visual, auditory, tactile, and other signal stimuli to produce their feelings. The features of virtual reality are summarized into '3i'. They are immersive, interactive, and imaginative [23]. Compared with the traditional " viewing from the outside - in silico operation, "VR emphasizes human beings' dominant role in virtual environments. With the high-speed development of virtual reality technology, its connotation is also further broadened. Based on Virtual Reality (VR), the concepts of Augmented Reality (AR), Mixed Reality (MR) were extended[24]. Although virtual reality technology is more diversified, it is mainly applied to breast cancer patients in immersive experiences using headgear and computer games.

In recent years, VR has received increasing attention as a symptom and rehabilitation training approach to address pain reduction, fatigue, anxiety, depression, and cognitive dysfunction in different groups. It has been applied for physical rehabilitation management of some diseases, such as stroke, Parkinson's, rectal cancer, and prostatic cancer[25-28]. Moreover, it has indeed received good results, and the quality of life of these patients has been dramatically improved. Considering that breast cancer patients have similar physical and mental burdens, it is feasible to attempt virtual reality into them. As for breast cancer patients, virtual reality technology was gradually introduced into American scholars' clinical intervention research in the early 21st century[28]. However, there are few intervention studies for breast cancer patients. Many of them are still in clinical trials, and there is no systematic review to summarize and analyze them. There is no specific intervention strategy. Existing systematic reviews primarily focused on the inclusion and analysis of mixed samples of cancer patients[29].

Therefore, we need to focus on specific populations of patients to be selected in research. Thus, doubts could be raised as to whether VR treatment methods for breast cancer can improve these symptoms; and whether the rehabilitation of patients is effective is worth studying. For this reason, we aimed to perform a meta-analysis as a scientific method to investigate the effects of this uncertain treatment using statistical methods to examine whether VR training is practical for breast cancer patients' symptoms and rehabilitation management. According to the current guidelines for breast cancer patients, we mainly focus on those who receive breast surgery or receive radiotherapy

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or chemotherapy on TNM 1-4 stage[30]. Based on the data analysis findings, we can make an objective evaluation and analysis of virtual reality technology in the rehabilitation and symptom management of breast cancer patients, hoping to provide a reference for the Application of patients and the clinical decision-making of medical staff.

2. Methods:

2.1 Search strategy

We checked up to three English databases (Pubmed, Embase, and CENTRAL), and one Chinese database (SinoMed) from the establishment of databases to November 2020, with the following Mesh terms and text words: ("Breast Neoplasms" OR "Breast Cancer Lymphedema" OR "breast tumor*" OR "breast cancer") AND ("Virtual Reality" OR "Virtual Reality Exposure Therapy" OR "simulation environment"). Appendix A shows the exact searching strategy. Finally, based on the above results, we performed a snowballing approach to search, screen, and examine papers, classifying them as potentially eligible studies.

2.2 Eligibility criteria

2.2.1 Types of studies:

We included all eligible randomized controlled trials, and quasi-experimental studies (published in English and Chinese), at least one of which was an ongoing virtual reality training program for inclusion in the review.

2.2.2 Types of participants:

Our inclusion population was required to meet the following criteria: i. breast cancer diagnosed by pathological analysis (TNM stage I–IV); ii. age ≥ 18 years old; iii. after surgery ongoing chemotherapy or radiotherapy. We also developed exclusion criteria for the participants: i. brain primary or metastatic disease (patients may have adverse

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reactions to VR); ii. combined with other severe diseases (such as severe heart disease, hypertension.); iii. previous psychiatric illness or cognitive impairment; iv. primary limb movement disorders; v. visual and auditory impairment.

2.2.3 Types of interventions:

The trial group participated in the organized and systematic intervention project based on virtual reality technology for breast cancer patients. The primary intervention goal was to reduce the psychological disorder, dysfunction, and pain of patients.

The intervention experiment based on virtual reality needs to meet the following core conditions:

1. Using computer-generated human-computer interactive intervention means in the virtual environment, such as VR glasses and headwear equipment;

2. Patients immerse themselves in the virtual environment constructed by electronic equipment;

3. Patients can perceive, feel, and interact in a manner that is similar to a physical place;

4. They can achieve effects by combining stimulation over multiple sensory channels such as vision, sound, touch, and perception to produce a corresponding stimulus.

And the potential comparison groups included any format of rehabilitation intervention, like exercise and usual care.

2.2.4 Types of outcomes:

Due to the different focus of the original research, the leading evaluation indicators may be various. To show the improvement of outcomes for patients by VR as comprehensively as possible and provide a reference for subsequent studies, we selected as many outcomes as possible under objective and fair evaluation conditions. The outcomes included: the patient's anxiety, fatigue, depressive symptoms, pain, cognitive function, and abduction of upper limbs. The measurement tools for outcomes are shown in the characteristics of the included studies.

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2.3 Data collection and analysis

2.3.1 Selection of studies

Two review authors independently screened all search results (title, abstract) in the database to identify studies that could be included in this paper. After initial screening, the review authors performed a full-text evaluation to determine the literature for inclusion. When disagreements occurred, they discussed the disparities and resolved them. If consensus was not reached after discussion between the two, it was decided by the third author independent arbitration. If necessary, we contacted study authors for specific information.

2.3.2 Data extraction and management

Two review authors independently extracted data onto a predesigned data collection form, including the trial setting, inclusion and exclusion criteria, study population, interventions, outcome measures, and final results of the papers. Both review authors, before data extraction, were given detailed instructions and participated in training sessions. Any disagreements or discrepancies regarding data extraction were resolved through discussion, and a third reviewer was consulted in case of any disagreement.

2.2.3 Assessment of risk of bias in included studies

We performed the risk of bias (ROB) of RCTs using Cochrane Collaboration ROB-2[31]. The overall assessment was recorded as high, low, and some concerns. Synthesis of ROB plots was done using online software Robvis (visualization tool)[32]. We used JBI critical appraisal checklist for quasi-experimental studies evaluated and included the before-after study[33].

2.3.4 Data analysis

All analyses were performed using RevMan V.5.4 (Cochrane Collaboration,

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http://ims.cochrane.org/revman). Heterogeneity was assessed using Cochran's Q test and quantified with the I^2-test. A random-effects model was applied to calculate the pooled results if $I^2 \geq 50\%$; otherwise, a fixed-effects model was used. Forest plot graphics were generated to present the pooled effect. The mean difference (MD) and standard MD (SMD) with corresponding 95% CI were used to calculate the effect size. SMD was calculated where the same outcome was reported but using different measurement tools. All tests are two-sided, and P-value<0.05 was considered to be statistically significant.

2.3.5 Dealing with missing data

There were no missing data of all these included articles.

3. Results:

3.1 Search process

After searching in 4 databases, we retrieved 3,058 articles. For these documents, we conducted a duplicate check, preliminary screening, and full-text reading for screening. Finally, eight papers met the criteria of this meta-analysis. 6 were randomized controlled trials, and 2 were before-after studies. Figure 1 shows the detailed search process and results.

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3.2 Critical appraisal of quality

Figure 2 shows the quality assessment of the included studies: 3 RCTs[34-36] judged by ROB-2 have a high risk of bias, 1[37] have a low risk of bias, while the remaining 2[40-41] studies showed the uncertain risk bias. The following were the primary sources of bias: deviations from intended intervention, differences in outcome measurement, and unclear randomization process. In summary, the quality of the included RCTs was considered as having a high risk of bias. 2[38,39] before-after studies judged by the JBI tool show the central bias exists in: comparison receiving similar care and multiple measurements of the outcome (Table 1).

Table1. Results of critical appraisal for quasi-experimental studies

Questions (potential bias and threat)	House et al. [38]	Chen et al. [39]
1. Is it clear in the study what is the 'cause' and what is the 'effect'?	Yes	Yes
2. Were the participants included in any comparisons similar?	Yes	Yes
3. Were the participants included in any comparisons receiving similar treatment/care, other than the exposure or intervention of interest?	No	Unclear
4. Was there a control group?	Yes	Yes
5. Were there multiple measurements of the outcome both pre and post the intervention/exposure?	Yes	No

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6. Was follow-up complete and if not, were differences between groups in terms of their follow-up adequately described and analyzed?	Yes	Yes
7. Were the outcomes of participants included in any comparisons measured in the same way?	Yes	Yes
8. Were outcomes measured in a reliable way?	Yes	Yes
9. Was appropriate statistical analysis used?	Yes	Yes

3.3 Study Characteristics

The essential characteristics of the included studies are shown in Table 2. The quality of the eight studies met our inclusion criteria. Three studies were from China [35,36,39], and the others were from the United States[38], Jordan[40], Italy[34], Turkey[37], and Australia[41]. Two of them were before -after controlled trials, and the other six were randomized controlled trials. These researches were conducted in the specialized ward and cancer center of the hospital. The number of participants ranged from 6 to 120. All participants were breast cancer patients who met the inclusion criteria. VR intervention included both immersive and non-immersive formats, and the duration of the intervention ranged from 20 minutes to 3 months. These studies assessed VR-based interventions' impact on health-related outcomes, including anxiety, depressive symptoms, fatigue, pain, cognitive function, and abduction of upper limbs.

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Table 2. Characteristics of the included studies

Study (Country/ year)	Design	Target population	Disease Stage /Age	Frequency (days/week)	Duration/ session (min)	Intervention time (weeks)	VR Method	Study Outcome	Study Results
House et al. [38] (USA/2016)	Before-after study	Post-surgical breast cancer women(n=6) with pain in the upper arm.	Unknown Age:57.8±20.4	2	20-50	8	The BrightArm Duo Rehabilitation System consists of a low-friction robotic rehabilitation table. A computerized forearm supports a display, a laptop computer for the therapist station, a remote clinical server, and a library of custom integrative rehabilitation games. It tracks arm position and grasping strength while patients play three-dimensional (3D) custom integrative rehabilitation games. Training difficulty increased progressively in game complexity, table tilt, and session length	BDI-II-depression symptom-decreased; BVMT-R/HVLT-R/TMT/NAB- cognitive function-improved; NRS-pain-decreased; Mechanical Goniometers-ROM-improved	PS PS PS PS
Chen et al. [39] (China/2019)	Before-after study	Breast cancer women (n=80) after surgery were hospitalized for chemotherapy.	II-III Age range:40-60	5	20-30	8	Patients with its motion-sensing controller and computer to build virtual cognitive rehabilitation training system, memory, executive ability, the training of the information processing speed and concentration of four modules, with the help of virtual scene and the virtual objects, through a variety of tasks such as puzzles, tai chi, a virtual maze game, improve patients' cognitive function, and real-time information feedback and rehabilitation information support.	MoCA -cognitive function-improved	PS
Mohammad et al. [40] (Jordan/2018)	RCT	Female patients with breast cancer(n=80) at a specialized cancer center	II-III Age:51.99±10.34	When giving morphine	15	When giving morphine	The CG (n=40) did not receive VR treatment. The VG(n=40) chose from two scenarios on a CD-ROM, which included deep-sea diving "Ocean Rift" or sitting on the beach with the "Happy Place" track. Then, the patients wore a head-mounted display with headphones. The PI remained near the participants during the VR session. The VR exposure session was ended at the peak time of painkiller efficacy.	VAS-pain-decreased; SAI-anxiety-decreased.	BS BS
Chirico et al. [34] (Italy/2019)	RCT	Female breast cancer patients(n=94) were receiving	I-III Age: VG:55.18±5.7 MTG:55.7±5.26	During each chemotherapy	20	During each chemotherapy	The CG(n=34) received routine care, and the MTG(n=30) received music therapy based on CG. The VG(n=30): The VR equipment consisted of a headset and an ahead movement tracking system. Participants explore the island through the relaxing landscape created	SAI-anxiety-decreased; SV-POMS(T/D/F)-depression symptom/fatigue-decreased	BS BS

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		chemotherapy.	CG:56.2±6.79				on the Second Life® platform through forests, animal observation, mountain climbing, and swimming.		
Feyzioğlu et al. [37] (Turkey/2020)	RCT	Breast cancer women(n=36) had undergone surgery.	Unknown; Age: VG:50.84±8.53; CG:51.00±7.06	2	45	6	The VG(n=19) received tissue massage, passive mobilization, and Xbox360Kinect™ video game treatment. The patient performed activities based on VR boxing, darts, and other games; the CG(n=17) received standard upper limb physical therapy treatment, including scar tissue massage and passive mobilization.	VAS-pain-decreased; Digital Goniometer-ROM-improved.	PS PS
Jimenez et al. [41] (Australia/2018)	RCT	Female breast cancer patients(n=37) received radiotherapy.	Unknown; Age range VG:35-74 CG: 35-74	Unknown	60	18	The VG(n=19) used the VERT system (Version 2.9) to provide basic technology, anatomical knowledge, and radiation dose information about RT for patients. The CG(n=18) received regular health education on radiotherapy knowledge.	STAI-S-anxiety-decreased.	PS
Jin a et al. [36] (China/2018)	RCT	Female breast cancer patients(n=120) undergoing surgery.	I-III Age range 20-50	6	30	12	The VG(n=60) used a professional rehabilitation platform; through the virtual and VR somatosensory equipment (helmet, joystick, data gloves), patients can choose their rehabilitation games for training. The CG(n=60) received traditional rehabilitation training.	SF-36-pain/fatigue-improved	BS
Jin b et al. [35] (China/2018)	RCT	Female breast cancer patients (n=76) received surgery.	II-III; Age range 33-69	14	15-30	12	The VG(n=38) selected appropriate training methods for rehabilitation according to the patients' muscle strength characteristics in different postoperative rehabilitation periods. The system includes sensors (HTC), upper limb rehabilitation devices, and a game platform (Unity 3D). The CG (n=38) received routine rehabilitation training under the guidance of nurses.	Mechanical Goniometers-ROM-improved	BS

Abbreviations: BDI-II: Beck Depression Inventory, Second Edition; BVMT-R: Brief Visuospatial Memory Test, Revised; CG: Control Group; HVLT-R: Hopkins Verbal Learning Test, Revised; MoCA: Montreal Cognitive Assessment; MTG: Music Therapy Group; NAB: Neuropsychological Assessment Battery; NRS: Numeric Pain Rating Scale; RCT: Randomized Controlled Trial; ROM: Range of Motion; SAI: State Anxiety Inventory; SF-36: 36-Item Short Form Survey; STAI-S: State-Trait Anxiety Inventory-State; SV-POMS(T/D/F): Short Version of Profile of Mood States in tense, depressive, and fatigue); TMT: Trail Making Test; VG: Virtual Reality Group; VAS: Visual Analog Scale.

BS: The comparison between the two groups was statistically significant.

PS: Outcome improved post-intervention, but intergroup results were not statistically significant, or there was no control group.

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3.4 Outcome analysis

Due to the limitation of research type and significant heterogeneity between groups, we only selected anxiety, fatigue, and abduction for meta-analysis. As for the remaining outcomes, we chose the form of qualitative analysis. We detailed the research form, time, equipment, and content of each study in Table 2. Although we could not conduct a meta-analysis, we also analyzed and expounded on their strengths and limitations.

3.4.1 Anxiety meta-analysis

Three studies[34,40,41] evaluated the impact of VR on breast cancer patients' anxiety. Compared with the control group, the results showed that VR-based intervention had a significant effect (SMD=-2.07, 95%CI= [-3.81 to -0.34], P=0.02, Figure 3). Heterogeneity was considerably high (I^2= 95%).

3.4.2 Abduction meta-analysis

Two studies[36,37] evaluated the impact of VR on breast cancer patients' abduction. Compared with the control group, the results showed that VR-based intervention had a significant effect (MD=15.54, 95%CI= [12.79 to 18.29], P<0.00001, Figure 4). Heterogeneity was considerably low (I^2= 0%).

3.4.3 Fatigue meta-analysis

Two studies[34,35] evaluated the impact of VR on breast cancer patients' fatigue. Compared with the control group, the results showed that VR-based intervention had no significant effect (SMD=-0.92, 95%CI= [-4.47 to 2.62], P=0.61, Figure 5).

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Heterogeneity was considerably high (I^2= 99%).

3.4.4 Depressive symptoms analysis

In two studies, participants were distracted by training in a recovery game in a virtual environment, and both before-after study[38] and randomized controlled trial[34] showed improvement in depressive symptoms. At the end of the intervention, the depression symptoms scores of patients in the before-after study decreased by 8.3 compared to baseline data($P<0.05$), and there were differences between the VR and control groups after the intervention in the randomized controlled trial($P<0.05$). The results were statistically significant.

3.4.5 Pain analysis

Three studies focused on patients' pain symptoms after the VR intervention, one before-after study[38], and two RCTs[37,40] showed a reduction in patients' pain scores. At the end of the intervention, patients' pain scores in the before-after study decreased by 1.5 compared to baseline($P<0.05$). There were differences in pain after intervention between the VR and control groups in two randomized controlled trials($P<0.05$). The results were statistically significant.

3.4.6 Cognitive function analysis

Two before-after studies[38,39] examined the effects of virtual reality on patients' cognitive function, showing significant improvements in executive function, language function, memory, and visual function. The results were statistically significant.

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4. Discussion

This systematic review and meta-analysis evaluated the application effect of intervention based-on virtual reality in breast cancer patients. Consistent with previous research[42], the results showed that compared with traditional breast cancer care, patients' anxiety, depressive symptoms, pain, cognitive function, and abduction of the upper limbs were improved with a statistical significance. However, the pooled effect showed that VR-based intervention had no effect on fatigue.

Our results showed that virtual reality had a significant effect on anxiety among patients with breast cancer. With the transformation from the traditional medical system to a biopsychosocial medical network, much more attention has been paid to the role of anxiety in the etiology and prognosis of breast cancer[43]. The Prevalence of anxiety among breast cancer patients was 41.9%[44]. The previous study assessed the effectiveness of VR for cancer patients, and their results showed VR had no significant effect on reducing anxiety in patients[29]. The difference in results might be that we focused on breast cancer patients, who were more uniform and had minor variations. Our results indicated that the heterogeneity of the included studies was significant. Due to the limited number of articles, we were unable to conduct sensitivity analysis to find the source of heterogeneity. According to the study design, we found that heterogeneity might exist in the specific treatment types of patients undergoing VR intervention, different pathological stages of cancer, small sample size, inconsistent equipment used, and different or unclear frequency and duration of intervention. Like anxiety, breast cancer patients usually get mental problems and physical symptoms. 97% of breast cancer patients report experiencing depression symptoms and pain. Our studies also

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included them in our studies; in each of the studies, the results showed that VR positively affected these symptoms. Because of the different types of research and lack of data, we could not show their combined effect sizes. We believe the possible mechanism behind VR's effect on the above symptoms is the distraction mechanism. According to Buhle et al. [45], the mechanism of distraction is different from placebo. They performed MRI analysis showed distraction effectively suppressed pain processing in the brain, while placebo may have no significant effect on pain processing. We hypothesize VR reduces pain through distraction, thereby improving patients' psychiatric symptoms during treatment. As mentioned before, our study was limited in many aspects, such as the type of study and sample, so we should treat this result with caution.

Our results showed that virtual reality had a significant effect on the abduction of upper limbs among patients with breast cancer. Bleeding, effusion, necrosis of skin flap, and upper limb edema often occur after radical breast cancer surgery, which will cause upper limb dysfunction such as limited shoulder joint activity and limb weakness, and significantly reduce the quality of life patients[46].VR as a rehabilitation tool has been very mature. Application in improving the range of motion of patients' upper limbs, especially in stroke patients, has been well verified[47]. Our results suggested that VR as a rehabilitation tool could improve some aspects of patients' upper limb movement. Still, we need more extensive sample size trials to confirm its long-term effects. VR may not be the only influential factor; it may need other auxiliary tools to best use its effects.

Cancer-related fatigue has been documented as one of the most distressing symptoms reported by breast cancer survivors. It affects functioning and impacts quality of life. Possible causal factors include physical conditions, affective and cognitive states,

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proinflammatory cytokines, and metabolic factors[48]. Our study found that virtual reality had no effect on fatigue among patients with breast cancer, which was contradictory to the findings of Zeng[29], who reported that VR-based intervention was related to a statistically significant decline in fatigue scores among patients with breast cancer. The possible reason for the discrepancy may be related to the differences in participants' characteristics. In addition, this may be related to the small sample size of literature included in the study, resulting in a significant bias. More strategies need to be explored to help breast cancer patients simultaneously manage the level of fatigue.

16% to 75% of breast cancer patients receiving chemotherapy have some degree of cognitive impairment[49]. Cognitive impairment seriously affects patients' quality of life, even more than the recurrence and metastasis of breast cancer itself. Two before-after studies examined the cognitive improvement of VR in breast cancer patients with improved vision, memory, executive function, and language function after the intervention. Due to technical reasons, we failed to conduct a combined effect size analysis, but VR is beneficial to the cognitive function of patients. The primary characteristic of VR is immersion, which enhances the subject's heart rate and activates brain circuit functions. In particular, it acts on the midline θ of the brain's forehead, including the frontal lobe and the left temporal area, bringing a pleasant emotional experience to subjects, activating the attention of related brain networks, and improving motor integration and spatial orientation functions[50].

To sum up, VR used the immersive experience to intervene and achieved positive results. These devices specifically included: wearable helmets, 3D glasses, somatosensory operating devices, smart displays, and computers. After these devices are connected, they can interact with the outside world in real-time. Subjects can

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interact cognitively or physically in a computer-generated virtual environment to achieve their expected results. An immersive tool changes patients' awareness in a virtual environment, provides a positive and pleasant experience, and improves intervention. Because of the interactivity of VR, patients enhance the effectiveness of an intervention in the real-time interaction of VR. This may be related to activating the midbrain marginal dopaminergic pathway and VR technology stimulation in the interaction between patients and VR[51]. The brain's reward mechanism can improve the attractiveness of participating in the intervention, enhance breast cancer patients' interest, and make patients more like this form. VR has real-time solid subjects in the virtual environment. Not only can real-time monitor the behavior and physiological response of patients, background data operation and guardians observe the transmission of stimulation in the virtual environment and comprehensively control the duration and intensity of rehabilitation[52]. The above utility mechanism improves the positive experience, attention, interaction, and participation of patients to a certain extent. It carries out rehabilitation in a real-time dynamic monitoring environment to improve patients' compliance, promote the rehabilitation of patients with breast cancer, and improve their prognosis.

Direction for future research and practice

The discussion shows VR for breast cancer symptom management and rehabilitation intervention methods are worth learning and promoting. With the rapid development of 5G communication technology, artificial intelligence, and Internet of Things technology, VR methods are bound to be more widely used and will receive technological innovation. Around the world, some countries have begun to formulate relevant laws and policies to give VR technical and economic support and policy

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inclination to accelerate industrial integration and put the new VR technology into clinical practice in medical and health care as soon as possible[53].

According to our research results, we state the following directions for further research. The Use of VR Security Question. When using a VR device, subjects are immersed in a virtual environment and often ignore the potential dangers in the actual scene. There is a risk of accidental injury and falling. Therefore, they should be monitored during use. At the same time, professionals should evaluate and modify their environment to ensure patient safety. Simultaneously, the articles retrieved in our systematic review rarely involve safety evaluation indicators, so we did not use them as outcome indicators. In the follow-up research, the safety evaluation system of VR application in patients can be constructed and applied. Be alert to the potential adverse effects of VR. The highly immersive nature of VR stimulates sensory nerves, which may lead to adverse reactions such as blurred vision, dizziness, nausea, and vomiting. Although the existing short-term intervention studies have not clearly defined it as an adverse event, long-term effect monitoring is still needed to avoid possible health problems and addiction caused by long-term use. The hardware and ecological construction are not yet mature. The Application of VR to breast cancer is still in its infancy. Most VR products on the market are not specifically developed for breast cancer patients. The clinical integration is low, ecological resource construction is not mature, and patients' targeted demands cannot be met. It can be low selectivity. It is suggested that a multidisciplinary team should be formed to develop VR hardware and ecological communities for breast cancer patients, build a more extensive resource library, enrich the functions of VR, and give patients more choices to maximize the benefits. During the COVID-19 pandemic, the social activities of the population were more alienated.

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This has demonstrated the importance of remote treatment systems when in-person patient-doctor contact is not possible. Because of the convenience and mobility of breast cancer intervention based on VR, patients can also be treated at home, reducing the risk of infection when going out and alleviating the feeling of social isolation. Telemedicine technology based on VR may become a trend to improve the utilization efficiency of medical resources.

Limitations

This study has some limitations that need to be addressed, and we had better treat the results of this study with caution. First of all, the types of studies included were relatively mixed, so we could not analyze the combined effect size in some outcomes. Although the qualitative analysis was carried out and the results were positive, we should treat it with caution because there was no quantitative analysis result. Secondly, we incorporated in the staging of breast cancer patients. However, most of the patients are in phase II-III; some patients' staging is unclear. The treatment of patients receiving also is not the same, so we can't be used for clinical population-specific recommendations, but we think VR has significant benefits for patients with mental symptoms. VR interventions included in the study are different in terms of intervention time, frequency, and specific intervention form, so there is heterogeneity. The small sample size and number of participants in our study may lead to considerable differences between groups. And for this reason, we were unable to check for publication bias in this study. In the future, we suggest researchers conduct multi-center randomized controlled trials to seek higher-quality evidence to prove the effect of VR on breast cancer patients.

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5. Conclusion

The systematic review showed that VR-based interventions improved anxiety, depressive symptoms, pain, cognitive function, and abduction of upper limbs in breast cancer patients, but not fatigue. Due to this paper's limitations and extensive sources, we were unable to carry out specific recommendations and conclusions. In the future, large-scale pilot studies on symptom management and rehabilitation in breast cancer patients should be carried out to explore its applicability and feasibility in breast cancer patients.

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Figure legend/caption: Figure1 Flow chart for study selection according to PRISMA Declaration 2020. PRISMA, Preferred Reporting Items for Systematic Reviews and

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Meta-Analyses; Figure2 Quality assessment of the included studies; Figure3 Effect of VR-based interventions on anxiety; Figure4 Effect of VR-based interventions on the abduction of upper limbs; Figure5 Effect of VR-based interventions on fatigue.

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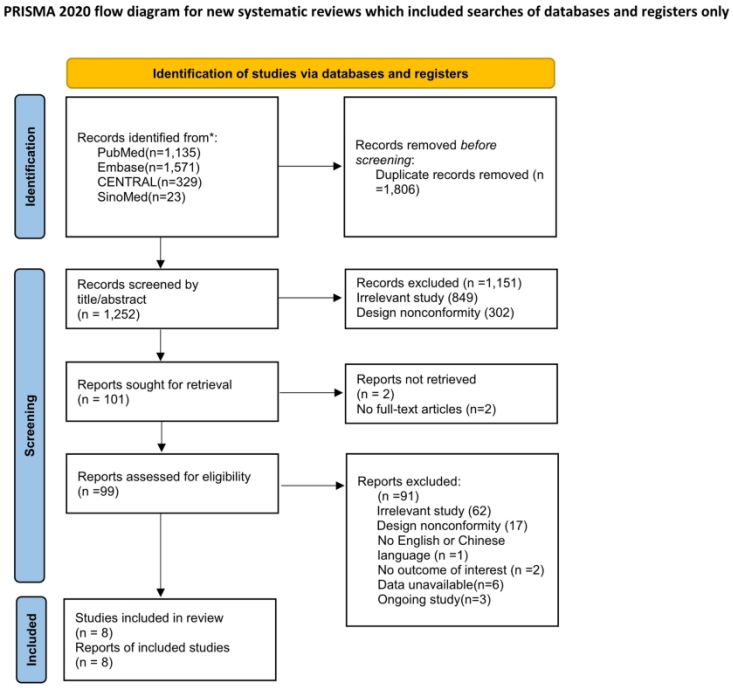


Figure1 Flow chart for study selection according to PRISMA Declaration 2020. PRISMA, Preferred Reporting Items for Systematic Reviews and Meta-Analyses

209x296mm (300 x 300 DPI)

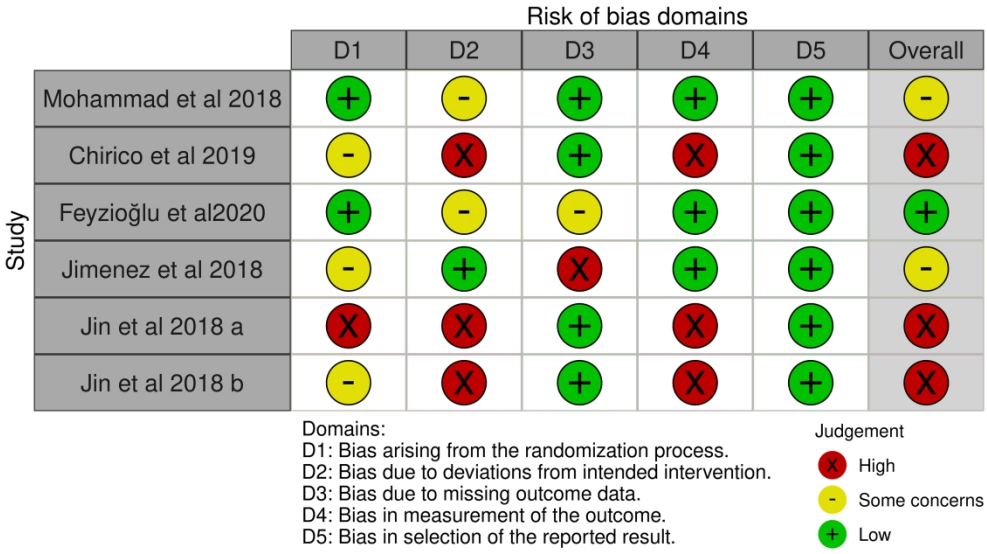


Figure2 Quality assessment of the included studies

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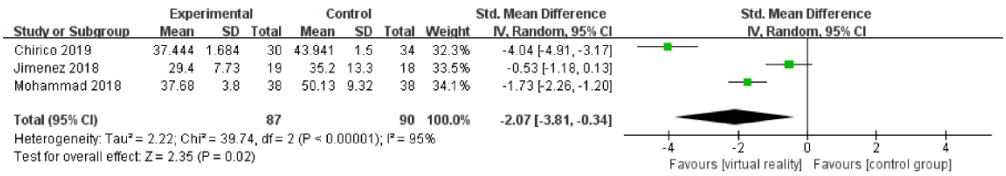


Figure3 Effect of VR-based interventions on anxiety

810x143mm (157 x 157 DPI)

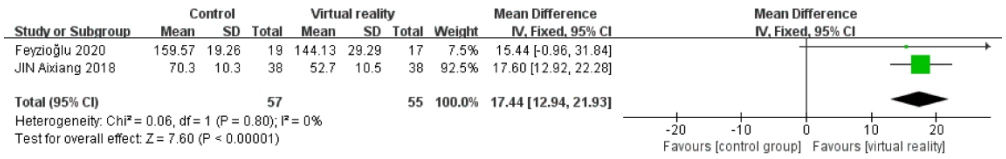


Figure4 Effect of VR-based interventions on the abduction of upper limbs

809x129mm (157 x 157 DPI)

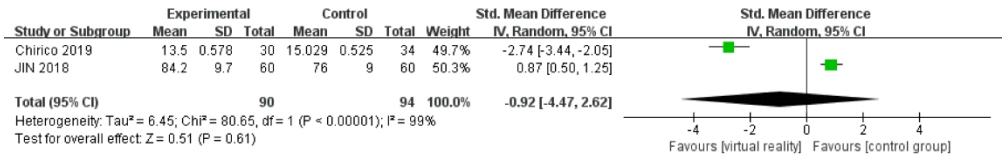


Figure5 Effect of VR-based interventions on fatigue

810x129mm (157 x 157 DPI)

Search Strategy

Appendix 1. PubMed (English)

1. Breast Neoplasms [MeSH Terms]
2. Breast Cancer Lymphedema [MeSH Terms]
3. breast tumor* [Title/Abstract]
4. breast cancer [Title/Abstract]
5. breast carcinoma* [Title/Abstract]
6. mammary gland carcinoma* [Title/Abstract]
7. Virtual Reality [MeSH Terms]
8. Virtual Reality Exposure Therapy [MeSH Terms]
9. User-Computer Interface [MeSH Terms]
10. virtual rehabilitation system [Title/Abstract]
11. virtual reality head-mounted display [Title/Abstract]
12. VR [Title/Abstract]
13. game [Title/Abstract]
14. gaming [Title/Abstract]
15. simulation system [Title/Abstract]
16. simulation environment [Title/Abstract]
17. 1 OR 2 OR 3 OR 4 OR 5 OR 6
18. 7 OR 8 OR 9 OR 10 OR 11 OR 12 OR 13 OR 14 OR 15 OR 16
19. 17 AND 18

Appendix 2. Embase (English)

1. 'breast cancer'/exp
2. 'breast tumor'/exp
3. 'breast cancer-related lymphedema'/exp
4. 'breast neoplasm*' [ab,ti]
5. 'breast carcinoma*' [ab,ti]
6. 'mammary gland carcinoma*' [ab,ti]
7. 'virtual reality'/exp
8. 'virtual reality exposure therapy'/exp
9. 'computer interface'/exp
10. 'virtual reality head mounted display'/exp
11. 'virtual rehabilitation system'/exp
12. 'vr' [ab,ti]
13. 'user-computer interface' [ab,ti]
14. 'game' [ab,ti]
15. 'gaming' [ab,ti]
16. 'simulation system' [ab,ti]
17. 'simulation environment' [ab,ti]
18. 'interactive' [ab,ti]
19. 'reality system' [ab,ti]
20. 1 OR 2 OR 3 OR 4 OR 5 OR 6

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21. 7 OR 8 OR 9 OR 10 OR 11 OR 12 OR 13 OR 14 OR 15 OR 16 OR 17 OR 18 OR 19
22. 21 AND 22

Appendix 3. CENTRAL (English)

1. MeSH descriptor [[Breast Neoplasms] explode all trees
2. MeSH descriptor [[Breast Cancer Lymphedema] explode all trees
3. breast tumor* [ti,ab,kw]
4. breast cancer [ti,ab,kw]
5. breast carcinoma* [ti,ab,kw]
6. mammary gland carcinoma* [ti,ab,kw]
7. MeSH descriptor [Virtual Reality] explode all trees
8. MeSH descriptor [Virtual Reality Exposure Therapy] explode all trees
9. MeSH descriptor [User-Computer Interface] explode all trees
10. virtual rehabilitation system [ti,ab,kw]
11. virtual reality head-mounted display [ti,ab,kw]
12. VR [ti,ab,kw]
13. game [ti,ab,kw]
14. gaming [ti,ab,kw]
15. simulation system [ti,ab,kw]
16. simulation environment [ti,ab,kw]
17. Interactive [ti,ab,kw]
18. reality system [ti,ab,kw]
19. 1 OR 2 OR 3 OR 4 OR 5 OR 6
20. 7 OR 8 OR 9 OR 10 OR 11 OR 12 OR 13 OR 14 OR 15 OR 16 OR 17 OR 18
21. 19 AND 20

Appendix 4. SinoMed (Chinese)

1. Breast Cancer
2. Virtual Reality
3. 1 AND 2

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	2
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	4-5
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	5-6
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	6-7
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	6
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	6 and Appendix A
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	8
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	8
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	8
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	8
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	8
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	Table 2
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	9
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	8
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	9
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	9
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	9
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	No
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	9
Certainty	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	9

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
assessment			
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	9 and Figure 1
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	Figure 1
Study characteristics	17	Cite each included study and present its characteristics.	Table 2
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	Table 1 and Figure 2
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	19-20 and Figure 3 4 5
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	19-20
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	19-20
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	19-20
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	19-20
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	19
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	Table 1 and Figure 2
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	21-23
	23b	Discuss any limitations of the evidence included in the review.	26
	23c	Discuss any limitations of the review processes used.	26
	23d	Discuss implications of the results for practice, policy, and future research.	24-25
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	3
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	Not registered
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	Not registered
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	27
Competing interests	26	Declare any competing interests of review authors.	27
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	In the manuscript and appendix

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
			document

From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71

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The Efficacy of Virtual Reality-Based Interventions for Breast Cancer Patients Symptom and Rehabilitation Management: A Systematic Review and Meta-Analysis

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Primary Subject Heading:	Oncology
Secondary Subject Heading:	Complementary medicine, Rehabilitation medicine
Keywords:	Breast tumours < ONCOLOGY, Information technology < BIOTECHNOLOGY & BIOINFORMATICS, REHABILITATION MEDICINE

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The Efficacy of Virtual Reality-Based Interventions for Breast Cancer Patients Symptom and Rehabilitation Management: A Systematic Review and Meta-Analysis

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Abstract

Objectives To determine the effectiveness of VR-based intervention on the symptoms and rehabilitation management in patients with breast cancer.

Design Systematic review and meta-analysis.

Study selection We included all eligible randomized controlled trials (RCTs) and quasi-experimental studies (published in English and Chinese).

Participants Breast cancer patients (≥18 years) undergoing cancer treatment.

Interventions Any intervention administered to improve the symptoms and rehabilitation of breast cancer patients. The control group was given conventional care.

Outcomes All outcomes were as follows: pain, fatigue, anxiety, depressive symptoms, cognitive function, and range of motion of upper limb in breast cancer patients.

Data sources We searched PubMed, Embase, CENTRAL, and SinoMed, four electronic databases, covering the database establishment period to January 2022.

Review methods Two reviewers independently extracted content and data consistent with the prespecified framework and assessed risk bias. Random-effects meta-analysis was used to pool data across trials. Meta-analysis was performed using Review Manager V.5.4.

Results: A total of 8 studies met the eligibility criteria and were included in this study. The combined effect size showed that VR was positive for improving patients' anxiety(SMD=-2.07, 95%CI= [-3.81 to -0.34], I^2= 95%) and abduction of upper limbs(MD=15.54, 95%CI= [12.79 to 18.29], I^2= 0%), but fatigue(SMD=-0.92, 95%CI= [-4.47 to 2.62], I^2= 99%) was not. Qualitative analysis showed VR improved patients' depressive symptoms, pain and cognitive function.

Conclusions: Virtual reality technology has a good effect on breast cancer patients' symptoms and rehabilitation management, but the quality of evidence is low, and the sample size is small. To date, there are few intervention studies, therefore, giving precise recommendation or conclusion is difficult. We have a favorable view of this, and more clinical studies are needed in the future to improve the credibility of the results.

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Keywords virtual reality; breast cancer; meta-analysis

Strengths and limitations of this study

1. Unlike previous systematic reviews that focused on many types of cancer patients, this is the first review that aims at investigating the effectiveness of virtual reality in breast cancer patients to evaluate the relative safety and effectiveness of this new technology.
2. Due to the diversity of research design, we used the random effect model for analysis.
3. Faced with the COVID-19 pandemic period, the intervention based on virtual reality provides patients to undergo rehabilitation effectively at home and reduce the risk of infection.
4. High heterogeneity in some outcome measures due to different measurement tools, low quality of included studies, and small sample size limit the further promotion of the results, resulting in a risk of bias.
5. The quality of the included literature is low, the long-term effect on patients is still unclear, there is a lack of literature support with a higher level of evidence.

Systematic review protocol and registration None.

Patient and public involvement Not applicable.

Ethics Statement Not applicable.

1. Introduction:

Breast cancer is the most commonly occurring cancer in women[1]. There were over 2 million new cases and nearly 630 thousand deaths globally in 2018[2]. Moreover, according to the prediction, new breast cancer patients will increase by more than 46% by 2040[3]. In recent years, there has been a remarkable improvement in breast cancer-related survival rate and life quality[4], because of the accessibility to various treatment options and rehabilitation options, for instance, early breast cancer screening, chemotherapy radiotherapy, surgical therapy, and comprehensive treatment

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rehabilitation management[5-9]. However, the long-term breast cancer disease progression and the treatment have some adverse effects, which can cause some physical and psychological negative consequences[10]. These possible long-term adverse effects include somatic symptoms such as limited joint range of motion, pain, and psychological distress such as fatigue, anxiety, and depressive symptoms[11-13]. Simultaneously, these symptoms and negative emotions may further cause cognitive dysfunctions for the patient, leading to poor predictive outcomes and affecting all aspects of their life[14,15].

Some adjuvant treatments are used for rehabilitation management of breast cancer patients to improve symptoms, side effects, and the quality of life[16]. This process is often carried out through traditional physical therapy and occupational therapy to improve the patients' symptoms and perform rehabilitation management[17]. However, these traditional methods are time-consuming, cumbersome, and cost a lot, and the results often depend on the medical staff's actual capabilities[18]. Moreover, traditional rehabilitation therapy is primarily mechanical repetitive training, which lacks interest and motivation[19]. Patients often lose interest in this process, which affects the training effect of patients. So the limitations of traditional rehabilitation training prompted the emergence of new and effective methods.

With the rapid development of computer science, more and more modern technologies are applied to managing symptoms and rehabilitation of breast cancer patients, and it is continuously developing[20].

Virtual reality (VR) is mainly used as a brand new rehabilitation therapy for breast cancer patients. The concept of VR was first proposed in the 1960s and gradually rose at the end of the 20th century, and it is an artificial intelligence technology based on multidisciplinary cooperation[21].

It uses the computer as the core architecture, calculates and simulates a three-dimensional virtual environment, and combines with other somatosensory devices (such as somatosensory handle, sensing helmet, 3D glasses) to make up the gap between the virtual environment and reality[22].

It allows interactive feedback between the user and the virtual environment, providing

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the user with visual, auditory, tactile, and other signal stimuli to produce their feelings. The features of virtual reality are summarized into '3i'. They are immersive, interactive, and imaginative [23]. Compared with the traditional " viewing from the outside - in silico operation, "VR emphasizes human beings' dominant role in virtual environments. With the high-speed development of virtual reality technology, its connotation is also further broadened. Based on Virtual Reality (VR), the concepts of Augmented Reality (AR), Mixed Reality (MR) were extended[24]. Although virtual reality technology is more diversified, it is mainly applied to breast cancer patients in immersive experiences using headgear and computer games.

In recent years, VR has received increasing attention as a symptom and rehabilitation training approach to address pain reduction, fatigue, anxiety, depression, and cognitive dysfunction in different groups. It has been applied for physical rehabilitation management of some diseases, such as stroke, Parkinson's, rectal cancer, and prostatic cancer[25-28]. Moreover, it has indeed received good results, and the quality of life of these patients has been dramatically improved. Considering that breast cancer patients have similar physical and mental burdens, it is feasible to attempt virtual reality into them. As for breast cancer patients, virtual reality technology was gradually introduced into American scholars' clinical intervention research in the early 21st century[28]. However, there are few intervention studies for breast cancer patients. Many of them are still in clinical trials, and there is no systematic review to summarize and analyze them. There is no specific intervention strategy. Existing systematic reviews primarily focused on the inclusion and analysis of mixed samples of cancer patients[29].

Therefore, we need to focus on specific populations of patients to be selected in research. Thus, doubts could be raised as to whether VR treatment methods for breast cancer can improve these symptoms; and whether the rehabilitation of patients is effective is worth studying. For this reason, we aimed to perform a meta-analysis as a scientific method to investigate the effects of this uncertain treatment using statistical methods to examine whether VR training is practical for breast cancer patients' symptoms and rehabilitation management. According to the current guidelines for breast cancer patients, we mainly focus on those who receive breast surgery or receive radiotherapy

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or chemotherapy on TNM 1-4 stage[30]. Based on the data analysis findings, we can make an objective evaluation and analysis of virtual reality technology in the rehabilitation and symptom management of breast cancer patients, hoping to provide a reference for the Application of patients and the clinical decision-making of medical staff.

2. Methods:

2.1 Search strategy

We checked up to three English databases (Pubmed, Embase, and CENTRAL), and one Chinese database (SinoMed) from the establishment of databases to November 2020, with the following Mesh terms and text words: ("Breast Neoplasms" OR "Breast Cancer Lymphedema" OR "breast tumor*" OR "breast cancer") AND ("Virtual Reality" OR "Virtual Reality Exposure Therapy" OR "simulation environment"). Appendix A shows the exact searching strategy. Finally, based on the above results, we performed a snowballing approach to search, screen, and examine papers, classifying them as potentially eligible studies.

2.2 Eligibility criteria

2.2.1 Types of studies:

We included all eligible randomized controlled trials, and quasi-experimental studies (published in English and Chinese), at least one of which was an ongoing virtual reality training program for inclusion in the review.

2.2.2 Types of participants:

Our inclusion population was required to meet the following criteria: i. breast cancer diagnosed by pathological analysis (TNM stage I–IV); ii. age ≥ 18 years old; iii. after surgery ongoing chemotherapy or radiotherapy. We also developed exclusion criteria for the participants: i. brain primary (patients may have adverse reactions to VR); ii.

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combined with other severe diseases (such as severe heart disease, hypertension.); iii. previous psychiatric illness or cognitive impairment; iv. primary limb movement disorders; v. visual and auditory impairment.

2.2.3 Types of interventions:

The trial group participated in the organized and systematic intervention project based on virtual reality technology for breast cancer patients. The primary intervention goal was to reduce the psychological disorder, dysfunction, and pain of patients.

The intervention experiment based on virtual reality needs to meet the following core conditions:

1. Using computer-generated human-computer interactive intervention means in the virtual environment, such as VR glasses and headwear equipment;
2. Patients immerse themselves in the virtual environment constructed by electronic equipment;
3. Patients can perceive, feel, and interact in a manner that is similar to a physical place;
4. They can achieve effects by combining stimulation over multiple sensory channels such as vision, sound, touch, and perception to produce a corresponding stimulus.

And the potential comparison groups included any format of rehabilitation intervention, like exercise and usual care.

2.2.4 Types of outcomes:

Due to the different focus of the original research, the leading evaluation indicators may be various. To show the improvement of outcomes for patients by VR as comprehensively as possible and provide a reference for subsequent studies, we selected as many outcomes as possible under objective and fair evaluation conditions. The outcomes included: the patient's anxiety, fatigue, depressive symptoms, pain, cognitive function, and abduction of upper limbs. The measurement tools for outcomes are shown in the characteristics of the included studies.

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2.3 Data collection and analysis

2.3.1 Selection of studies

Two review authors independently screened all search results (title, abstract) in the database to identify studies that could be included in this paper. After initial screening, the review authors performed a full-text evaluation to determine the literature for inclusion. When disagreements occurred, they discussed the disparities and resolved them. If consensus was not reached after discussion between the two, it was decided by the third author independent arbitration. If necessary, we contacted study authors for specific information.

2.3.2 Data extraction and management

Two review authors independently extracted data onto a predesigned data collection form, including the trial setting, inclusion and exclusion criteria, study population, interventions, outcome measures, and final results of the papers. Both review authors, before data extraction, were given detailed instructions and participated in training sessions. Any disagreements or discrepancies regarding data extraction were resolved through discussion, and a third reviewer was consulted in case of any disagreement.

2.2.3 Assessment of risk of bias in included studies

We performed the risk of bias (ROB) of RCTs using Cochrane Collaboration ROB-2[31]. The overall assessment was recorded as high, low, and some concerns. Synthesis of ROB plots was done using online software Robvis (visualization tool)[32]. We used JBI critical appraisal checklist for quasi-experimental studies evaluated and included the before-after study[33].

2.3.4 Data analysis

All analyses were performed using RevMan V.5.4 (Cochrane Collaboration,

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http://ims.cochrane.org/revman). Heterogeneity was assessed using Cochran's Q test and quantified with the I^2-test. A random-effects model was applied to calculate the pooled results if $I^2 \geq 50\%$; otherwise, a fixed-effects model was used. Forest plot graphics were generated to present the pooled effect. The mean difference (MD) and standard MD (SMD) with corresponding 95% CI were used to calculate the effect size. SMD was calculated where the same outcome was reported but using different measurement tools. All tests are two-sided, and P-value<0.05 was considered to be statistically significant.

2.3.5 Dealing with missing data

There was no missing data for any of the included articles.

3. Results:

3.1 Search process

After searching in 4 databases, we retrieved 3,058 articles. For these documents, we conducted a duplicate check, preliminary screening, and full-text reading for screening. Finally, eight papers met the criteria of this meta-analysis. 6 were randomized controlled trials, and 2 were before-after studies. Figure 1 shows the detailed search process and results.

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3.2 Critical appraisal of quality

Figure 2 shows the quality assessment of the included studies: 3 RCTs[34-36] judged by ROB-2 have a high risk of bias, 1[37] have a low risk of bias, while the remaining 2[40-41] studies showed the uncertain risk bias. The following were the primary sources of bias: deviations from intended intervention, differences in outcome measurement, and unclear randomization process. In summary, the quality of the included RCTs was considered as having a high risk of bias. 2[38,39] before-after studies judged by the JBI tool show the central bias exists in: comparison receiving similar care and multiple measurements of the outcome (Table 1).

Table1. Results of critical appraisal for quasi-experimental studies

Questions (potential bias and threat)	House et al. [38]	Chen et al. [39]
1. Is it clear in the study what is the 'cause' and what is the 'effect'?	Yes	Yes
2. Were the participants included in any comparisons similar?	Yes	Yes
3. Were the participants included in any comparisons receiving similar treatment/care, other than the exposure or intervention of interest?	No	Unclear
4. Was there a control group?	Yes	Yes
5. Were there multiple measurements of the outcome both pre and post the intervention/exposure?	Yes	No

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6. Was follow-up complete and if not, were differences between groups in terms of their follow-up adequately described and analyzed?	Yes	Yes
7. Were the outcomes of participants included in any comparisons measured in the same way?	Yes	Yes
8. Were outcomes measured in a reliable way?	Yes	Yes
9. Was appropriate statistical analysis used?	Yes	Yes

3.3 Study Characteristics

The essential characteristics of the included studies are shown in Table 2. The quality of the eight studies met our inclusion criteria. Three studies were from China [35,36,39], and the others were from the United States[38], Jordan[40], Italy[34], Turkey[37], and Australia[41]. Two of them were before -after controlled trials, and the other six were randomized controlled trials. The research was conducted in specialized wards and cancer centers of hospitals. The number of participants ranged from 6 to 120. All participants were breast cancer patients who met the inclusion criteria. VR intervention included both immersive and non-immersive formats, and the duration of the intervention ranged from 20 minutes to 3 months. The studies assessed VR-based interventions' impact on health-related outcomes, including anxiety, depressive symptoms, fatigue, pain, cognitive function, and abduction of upper limbs.

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Table 2. Characteristics of the included studies

Study (Country/ year)	Design	Target population	Disease Stage /Age	Frequency (days/week)	Duration/ session (min)	Intervention time (weeks)	VR Method	Study Outcome	Study Results
House et al. [38] (USA/2016)	Before-after study	Post-surgical breast cancer women(n=6) with pain in the upper arm.	Unknown Age:57.8±20.4	2	20-50	8	The BrightArm Duo Rehabilitation System consists of a low-friction robotic rehabilitation table. A computerized forearm supports a display, a laptop computer for the therapist station, a remote clinical server, and a library of custom integrative rehabilitation games. It tracks arm position and grasping strength while patients play three-dimensional (3D) custom integrative rehabilitation games. Training difficulty increased progressively in game complexity, table tilt, and session length	BDI-II-depression symptom-decreased; BVMT-R/HVLT-R/TMT/NAB- cognitive function-improved; NRS-pain-decreased; Mechanical Goniometers-ROM-improved	PS PS PS PS
Chen et al. [39] (China/2019)	Before-after study	Breast cancer women (n=80) after surgery were hospitalized for chemotherapy.	II-III Age range:40-60	5	20-30	8	Patients with its motion-sensing controller and computer to build virtual cognitive rehabilitation training system, memory, executive ability, the training of the information processing speed and concentration of four modules, with the help of virtual scene and the virtual objects, through a variety of tasks such as puzzles, tai chi, a virtual maze game, improve patients' cognitive function, and real-time information feedback and rehabilitation information support.	MoCA -cognitive function-improved	PS
Mohammad et al. [40] (Jordan/2018)	RCT	Female patients with breast cancer(n=80) at a specialized cancer center	II-III Age:51.99±10.34	When giving morphine	15	When giving morphine	The CG (n=40) did not receive VR treatment. The VG(n=40) chose from two scenarios on a CD-ROM, which included deep-sea diving "Ocean Rift" or sitting on the beach with the "Happy Place" track. Then, the patients wore a head-mounted display with headphones. The PI remained near the participants during the VR session. The VR exposure session was ended at the peak time of painkiller efficacy.	VAS-pain-decreased; SAI-anxiety-decreased.	BS BS
Chirico et al. [34] (Italy/2019)	RCT	Female breast cancer patients(n=94) were receiving	I-III Age: VG:55.18±5.7 MTG:55.7±5.26	During each chemotherapy	20	During each chemotherapy	The CG(n=34) received routine care, and the MTG(n=30) received music therapy based on CG. The VG(n=30): The VR equipment consisted of a headset and an ahead movement tracking system. Participants explore the island through the relaxing landscape created	SAI-anxiety-decreased; SV-POMS(T/D/F)-depression symptom/fatigue-decreased	BS BS

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		chemotherapy.	CG:56.2±6.79				on the Second Life® platform through forests, animal observation, mountain climbing, and swimming.		
Feyzioğlu et al. [37] (Turkey/2020)	RCT	Breast cancer women(n=36) had undergone surgery.	Unknown; Age: VG:50.84±8.53; CG:51.00±7.06	2	45	6	The VG(n=19) received tissue massage, passive mobilization, and Xbox360Kinect™ video game treatment. The patient performed activities based on VR boxing, darts, and other games; the CG(n=17) received standard upper limb physical therapy treatment, including scar tissue massage and passive mobilization.	VAS-pain-decreased; Digital Goniometer-ROM-improved.	PS PS
Jimenez et al. [41] (Australia/2018)	RCT	Female breast cancer patients(n=37) received radiotherapy.	Unknown; Age range VG:35-74 CG: 35-74	Unknown	60	18	The VG(n=19) used the VERT system (Version 2.9) to provide basic technology, anatomical knowledge, and radiation dose information about RT for patients. The CG(n=18) received regular health education on radiotherapy knowledge.	STAI-S-anxiety-decreased.	PS
Jin a et al. [36] (China/2018)	RCT	Female breast cancer patients(n=120) undergoing surgery.	I-III Age range 20-50	6	30	12	The VG(n=60) used a professional rehabilitation platform; through the virtual and VR somatosensory equipment (helmet, joystick, data gloves), patients can choose their rehabilitation games for training. The CG(n=60) received traditional rehabilitation training.	SF-36-pain/fatigue-improved	BS
Jin b et al. [35] (China/2018)	RCT	Female breast cancer patients (n=76) received surgery.	II-III; Age range 33-69	14	15-30	12	The VG(n=38) selected appropriate training methods for rehabilitation according to the patients' muscle strength characteristics in different postoperative rehabilitation periods. The system includes sensors (HTC), upper limb rehabilitation devices, and a game platform (Unity 3D). The CG (n=38) received routine rehabilitation training under the guidance of nurses.	Mechanical Goniometers-ROM-improved	BS

Abbreviations: BDI-II: Beck Depression Inventory, Second Edition; BVMT-R: Brief Visuospatial Memory Test, Revised; CG: Control Group; HVLT-R: Hopkins Verbal Learning Test, Revised; MoCA: Montreal Cognitive Assessment; MTG: Music Therapy Group; NAB: Neuropsychological Assessment Battery; NRS: Numeric Pain Rating Scale; RCT: Randomized Controlled Trial; ROM: Range of Motion; SAI: State Anxiety Inventory; SF-36: 36-Item Short Form Survey; STAI-S: State-Trait Anxiety Inventory-State; SV-POMS(T/D/F): Short Version of Profile of Mood States in tense, depressive, and fatigue); TMT: Trail Making Test; VG: Virtual Reality Group; VAS: Visual Analog Scale.

BS: The comparison between the two groups was statistically significant.

PS: Outcome improved post-intervention, but intergroup results were not statistically significant, or there was no control group.

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3.4 Outcome analysis

Due to the limitation of research type and significant heterogeneity between groups, we only selected anxiety, fatigue, and abduction for meta-analysis. As for the remaining outcomes, we chose the form of qualitative analysis. We detailed the research form, time, equipment, and content of each study in Table 2. Although we could not conduct a meta-analysis, we also analyzed and expounded on their strengths and limitations.

3.4.1 Anxiety meta-analysis

Three studies[34,40,41] evaluated the impact of VR on breast cancer patients' anxiety. Compared with the control group, the results showed that VR-based intervention had a significant effect (SMD=-2.07, 95%CI= [-3.81 to -0.34], P=0.02, Figure 3). Heterogeneity was considerably high (I^2= 95%).

3.4.2 Abduction meta-analysis

Two studies[36,37] evaluated the impact of VR on breast cancer patients' abduction. Compared with the control group, the results showed that VR-based intervention had a significant effect (MD=15.54, 95%CI= [12.79 to 18.29], P<0.00001, Figure 4). Heterogeneity was considerably low (I^2= 0%).

3.4.3 Fatigue meta-analysis

Two studies[34,35] evaluated the impact of VR on breast cancer patients' fatigue. Compared with the control group, the results showed that VR-based intervention had no significant effect (SMD=-0.92, 95%CI= [-4.47 to 2.62], P=0.61, Figure 5).

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Heterogeneity was considerably high (I^2= 99%).

3.4.4 Depressive symptoms analysis

In two studies, participants were distracted by training in a recovery game in a virtual environment, and both before-after study[38] and randomized controlled trial[34] showed improvement in depressive symptoms. At the end of the intervention, the depression symptoms scores (MD) of patients in the before-after study decreased by 8.3 compared to baseline data($P<0.05$), and there were differences between the VR and control groups after the intervention in the randomized controlled trial($P<0.05$). The results were statistically significant.

3.4.5 Pain analysis

Three studies focused on patients' pain symptoms after the VR intervention, one before-after study[38], and two RCTs[37,40] showed a reduction in patients' pain scores. At the end of the intervention, patients' pain scores (MD) in the before-after study decreased by 1.5 compared to baseline($P<0.05$). There were differences in pain after intervention between the VR and control groups in two randomized controlled trials($P<0.05$). The results were statistically significant.

3.4.6 Cognitive function analysis

Two before-after studies[38,39] examined the effects of virtual reality on patients' cognitive function, showing significant improvements in executive function, language function, memory, and visual function. The results were statistically significant.

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4. Discussion

This systematic review and meta-analysis evaluated the application effect of intervention based-on virtual reality in breast cancer patients. Consistent with previous research[42], the results showed that compared with traditional breast cancer care, patients' anxiety, depressive symptoms, pain, cognitive function, and abduction of the upper limbs were improved with a statistical significance. However, the pooled effect showed that VR-based intervention had no effect on fatigue.

Our results showed that virtual reality had a significant effect on anxiety among patients with breast cancer. With the transformation from the traditional medical system to a biopsychosocial medical network, much more attention has been paid to the role of anxiety in the etiology and prognosis of breast cancer[43]. The Prevalence of anxiety among breast cancer patients was 41.9%[44]. The previous study assessed the effectiveness of VR for cancer patients, and their results showed VR had no significant effect on reducing anxiety in patients[29]. The difference in results might be that we focused on breast cancer patients, who were more uniform and had minor variations. Our results indicated that the heterogeneity of the included studies was significant. Due to the limited number of articles, we were unable to conduct sensitivity analysis to find the source of heterogeneity. According to the study design, we found that heterogeneity might exist in the specific treatment types of patients undergoing VR intervention, different pathological stages of cancer, small sample size, inconsistent equipment used, and different or unclear frequency and duration of intervention. Like anxiety, breast cancer patients usually get mental problems and physical symptoms. 97% of breast cancer patients report experiencing depression symptoms and pain. Our review also

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included the effect of VR on depressive symptoms. In both studies analyzed, the VR treatment had a reducing effect on depressive symptoms. Because of the different types of research and lack of data, we could not show their combined effect sizes. We believe the possible mechanism behind VR's effect on the above symptoms is the distraction mechanism. According to Buhle et al. [45], the mechanism of distraction is different from placebo. Their MRI analysis showed distraction effectively suppressed pain processing in the brain, while placebo had no significant effect on pain processing. We hypothesize VR reduces pain through distraction, thereby improving patients' psychiatric symptoms during treatment. As mentioned before, our study was limited in many aspects, such as the type of study and sample, so we should treat this result with caution.

Our results showed that virtual reality had a significant effect on the abduction of upper limbs among patients with breast cancer. Bleeding, effusion, necrosis of skin flap, and upper limb edema often occur after radical breast cancer surgery, which will cause upper limb dysfunction such as limited shoulder joint activity and limb weakness, and significantly reduce the quality of life of patients[46].VR as a rehabilitation tool has been very mature. Application in improving the range of motion of patients' upper limbs, especially in stroke patients, has been well verified[47]. Our results suggested that VR as a rehabilitation tool could improve some aspects of patients' upper limb movement. Still, we need more extensive sample size trials to confirm its long-term effects. VR may not be the only influential factor; it may need other auxiliary tools to best use its effects.

Cancer-related fatigue has been documented as one of the most distressing symptoms reported by breast cancer survivors. It affects functioning and impacts quality of life. Possible causal factors include physical conditions, affective and cognitive states, proinflammatory cytokines, and metabolic factors[48]. Our study found that virtual reality

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had no effect on fatigue among patients with breast cancer, which was contradictory to the findings of Zeng[29], who reported that VR-based intervention was related to a statistically significant decline in fatigue scores among patients with breast cancer. The possible reason for the discrepancy may be related to the differences in participants' characteristics. In addition, this may be related to the small sample size of literature included in the study, resulting in a significant bias. More strategies need to be explored to help breast cancer patients simultaneously manage the level of fatigue.

16% to 75% of breast cancer patients receiving chemotherapy have some degree of cognitive impairment[49]. Cognitive impairment seriously affects patients' quality of life, even more than the recurrence and metastasis of breast cancer itself. Two before-after studies examined the cognitive improvement of VR in breast cancer patients with improved vision, memory, executive function, and language function after the intervention. Due to technical reasons, we failed to conduct a combined effect size analysis, but from the data analysis of the before-after study, VR is beneficial to the cognitive function of patients. The primary characteristic of VR is immersion, which enhances the subject's heart rate and activates brain circuit functions. In particular, it acts on the midline θ of the brain's forehead, including the frontal lobe and the left temporal area, bringing a pleasant emotional experience to subjects, activating the attention of related brain networks, and improving motor integration and spatial orientation functions[50].

To sum up, in this context VR uses immersive experience to intervene and achieve positive effects for breast cancer patients. The devices which were used are: wearable helmets, 3D glasses, somatosensory operating devices, smart displays, and computers. Subjects can interact cognitively or physically in a computer-generated virtual

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environment in real-time to achieve their goals. An immersive tool changes patients' focus of attention, provides a positive and pleasant experience, and improves progress. Because of the interactivity of VR, patients enhance the effectiveness of an intervention in the real-time interaction of VR. This may be related to activating the midbrain marginal dopaminergic pathway and VR technology stimulation in the interaction between patients and VR[51]. The brain's reward mechanism can improve the attractiveness of participating in the intervention, enhance breast cancer patients' interest, and make patients more like this form. VR has the characteristics of real-time in virtual environment. Not only can real-time monitor the behavior and physiological response of patients, background data operation and guardians observe the transmission of stimulation in the virtual environment and comprehensively control the duration and intensity of rehabilitation[52]. The above utility mechanism improves the positive experience, attention, interaction, and participation of patients to a certain extent. It carries out rehabilitation in a real-time dynamic monitoring environment to improve patients' compliance, promote the rehabilitation of patients with breast cancer, and improve their prognosis.

Direction for future research and practice

The discussion shows VR for breast cancer symptom management and rehabilitation intervention methods are worth learning and promoting. With the rapid development of 5G communication technology, artificial intelligence, and Internet of Things technology, VR methods are bound to be more widely used and will receive technological innovation. Around the world, some countries have begun to formulate relevant laws and policies to give VR technical and economic support and policy inclination to accelerate industrial integration and put the new VR technology into

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clinical practice in medical and health care as soon as possible[53].

According to our research results, we state the following directions for further research. The Use of VR Security Question. When using a VR device, subjects are immersed in a virtual environment and often ignore the potential dangers in the actual scene. There is a risk of accidental injury and falling. Therefore, they should be monitored during use. At the same time, professionals should evaluate and modify their environment to ensure patient safety. Simultaneously, the articles retrieved in our systematic review rarely involve safety evaluation indicators, so we did not use them as outcome indicators. In the follow-up research, the safety evaluation system of VR application in patients can be constructed and applied. Be alert to the potential adverse effects of VR. The highly immersive nature of VR stimulates sensory nerves, which may lead to adverse reactions such as blurred vision, dizziness, nausea, and vomiting. Although the existing short-term intervention studies have not clearly defined it as an adverse event, long-term effect monitoring is still needed to avoid possible health problems and addiction caused by long-term use. The hardware and ecological construction are not yet mature. The Application of VR to breast cancer is still in its infancy. Most VR products on the market are not specifically developed for breast cancer patients. The clinical integration is low, ecological resource construction is not mature, and patients' targeted demands cannot be met. It can be low selectivity. It is suggested that a multidisciplinary team should be formed to develop VR hardware and ecological communities for breast cancer patients, build a more extensive resource library, enrich the functions of VR, and give patients more choices to maximize the benefits. During the COVID-19 pandemic, the social activities of the population were more alienated. This has demonstrated the importance of remote treatment systems when in-person

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patient-doctor contact is not possible. Because of the convenience and mobility of breast cancer intervention based on VR, patients can also be treated at home, reducing the risk of infection when going out and alleviating the feeling of social isolation. Telemedicine technology based on VR may become a trend to improve the utilization efficiency of medical resources.

Limitations

This study has some limitations that need to be addressed, and which is why we need to interpret the results with caution. First of all, the types of studies included were relatively diverse, so we could not analyze the combined effect size in some outcomes. Although the qualitative analysis was carried out and the results were positive, we should treat it with caution because there was no quantitative analysis result. Secondly, we clarified and discussed the staging of breast cancer patients. Most patients were in stage II-III, and the staging of patients in some of the included studies was unclear. The specific form of VR therapy that patients received also varies, so we cannot recommend its use for specific clinical populations, but based on our findings, we believe VR has significant benefits for patients with mental symptoms. VR interventions included in the study are different in terms of intervention time, frequency, and specific intervention form, so there is heterogeneity. The small sample size and number of participants in our study may lead to considerable differences between groups. And for this reason, we were unable to check for publication bias in this study. In the future, we suggest researchers conduct multi-center randomized controlled trials to seek higher-quality evidence to prove the effect of VR on breast cancer patients.

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5. Conclusion

The systematic review showed that VR-based interventions improved anxiety, depressive symptoms, pain, cognitive function, and abduction of upper limbs in breast cancer patients, but not fatigue. Due to this paper's limitations and extensive sources, we were unable to carry out specific recommendations and conclusions. In the future, large-scale pilot studies on symptom management and rehabilitation in breast cancer patients should be carried out to explore its applicability and feasibility in breast cancer patients.

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Figure legend/caption: Figure1 Flow chart for study selection according to PRISMA Declaration 2020. PRISMA, Preferred Reporting Items for Systematic Reviews and

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Meta-Analyses; Figure2 Quality assessment of the included studies; Figure3 Effect of VR-based interventions on anxiety; Figure4 Effect of VR-based interventions on the abduction of upper limbs; Figure5 Effect of VR-based interventions on fatigue.

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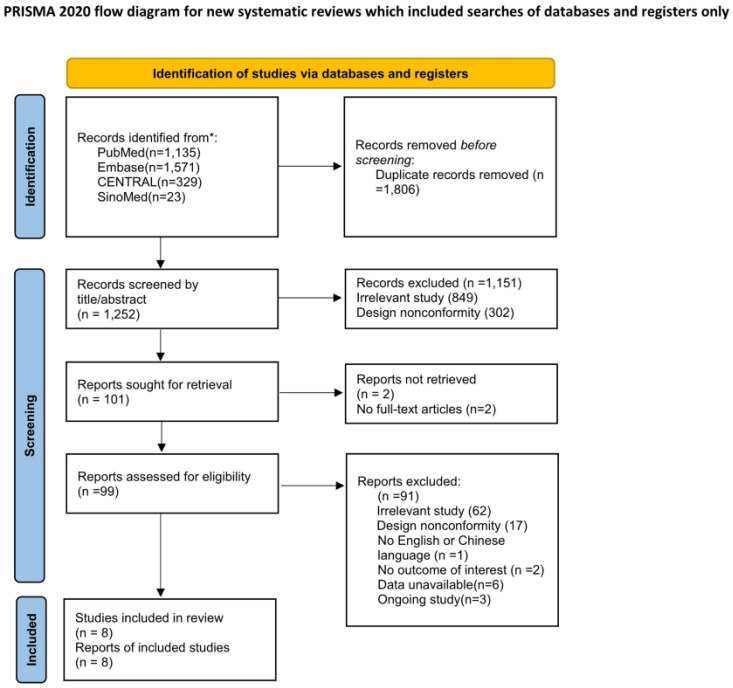


Figure1 Flow chart for study selection according to PRISMA Declaration 2020. PRISMA, Preferred Reporting Items for Systematic Reviews and Meta-Analyses

210x296mm (500 x 500 DPI)

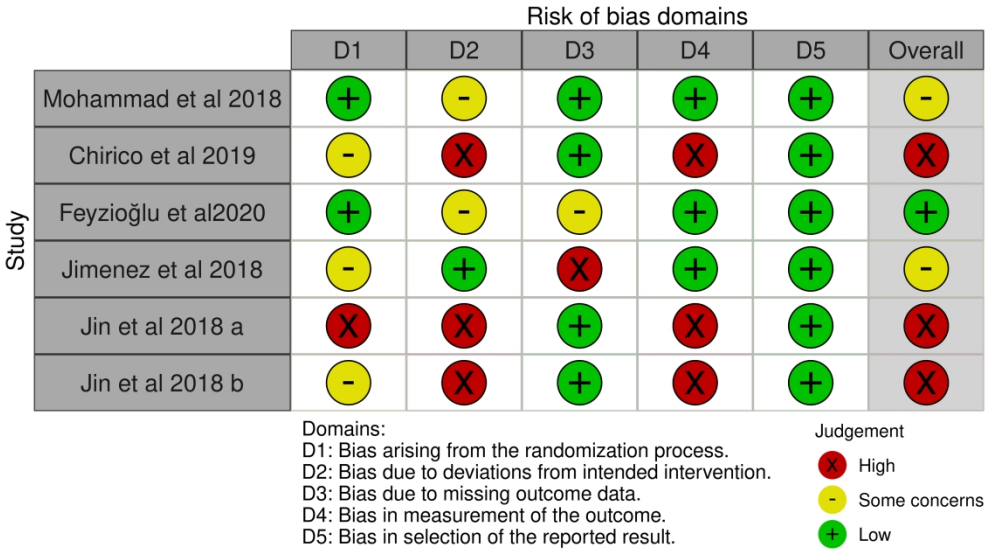


Figure2 Quality assessment of the included studies

447x283mm (197 x 197 DPI)

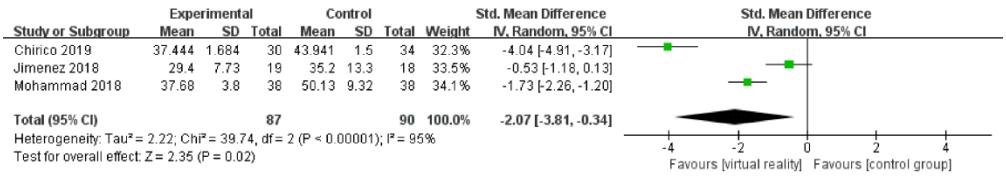


Figure3 Effect of VR-based interventions on anxiety

810x143mm (157 x 157 DPI)

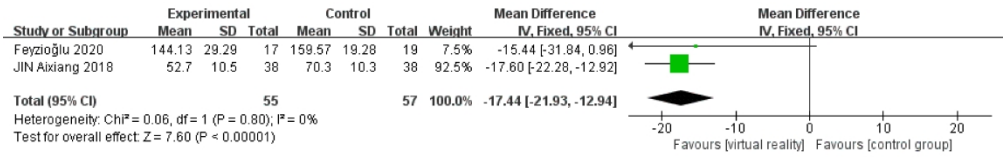


Figure4 Effect of VR-based interventions on the abduction of upper limbs

820x129mm (157 x 157 DPI)

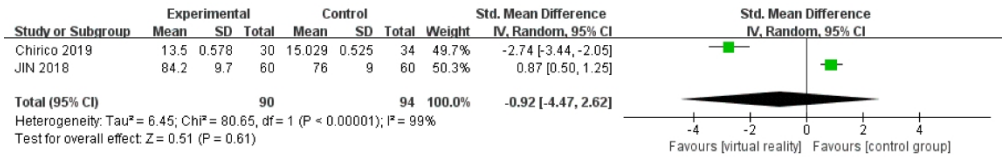


Figure5 Effect of VR-based interventions on fatigue

810x129mm (157 x 157 DPI)

Search Strategy

Appendix 1. PubMed (English)

1. Breast Neoplasms [MeSH Terms]
2. Breast Cancer Lymphedema [MeSH Terms]
3. breast tumor* [Title/Abstract]
4. breast cancer [Title/Abstract]
5. breast carcinoma* [Title/Abstract]
6. mammary gland carcinoma* [Title/Abstract]
7. Virtual Reality [MeSH Terms]
8. Virtual Reality Exposure Therapy [MeSH Terms]
9. User-Computer Interface [MeSH Terms]
10. virtual rehabilitation system [Title/Abstract]
11. virtual reality head-mounted display [Title/Abstract]
12. VR [Title/Abstract]
13. game [Title/Abstract]
14. gaming [Title/Abstract]
15. simulation system [Title/Abstract]
16. simulation environment [Title/Abstract]
17. 1 OR 2 OR 3 OR 4 OR 5 OR 6
18. 7 OR 8 OR 9 OR 10 OR 11 OR 12 OR 13 OR 14 OR 15 OR 16
19. 17 AND 18

Appendix 2. Embase (English)

1. 'breast cancer'/exp
2. 'breast tumor'/exp
3. 'breast cancer-related lymphedema'/exp
4. 'breast neoplasm*' [ab,ti]
5. 'breast carcinoma*' [ab,ti]
6. 'mammary gland carcinoma*' [ab,ti]
7. 'virtual reality'/exp
8. 'virtual reality exposure therapy'/exp
9. 'computer interface'/exp
10. 'virtual reality head mounted display'/exp
11. 'virtual rehabilitation system'/exp
12. 'vr' [ab,ti]
13. 'user-computer interface' [ab,ti]
14. 'game' [ab,ti]
15. 'gaming' [ab,ti]
16. 'simulation system' [ab,ti]
17. 'simulation environment' [ab,ti]
18. 'interactive' [ab,ti]
19. 'reality system' [ab,ti]
20. 1 OR 2 OR 3 OR 4 OR 5 OR 6

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21. 7 OR 8 OR 9 OR 10 OR 11 OR 12 OR 13 OR 14 OR 15 OR 16 OR 17 OR 18 OR 19
22. 21 AND 22

Appendix 3. CENTRAL (English)

1. MeSH descriptor [[Breast Neoplasms] explode all trees
2. MeSH descriptor [[Breast Cancer Lymphedema] explode all trees
3. breast tumor* [ti,ab,kw]
4. breast cancer [ti,ab,kw]
5. breast carcinoma* [ti,ab,kw]
6. mammary gland carcinoma* [ti,ab,kw]
7. MeSH descriptor [Virtual Reality] explode all trees
8. MeSH descriptor [Virtual Reality Exposure Therapy] explode all trees
9. MeSH descriptor [User-Computer Interface] explode all trees
10. virtual rehabilitation system [ti,ab,kw]
11. virtual reality head-mounted display [ti,ab,kw]
12. VR [ti,ab,kw]
13. game [ti,ab,kw]
14. gaming [ti,ab,kw]
15. simulation system [ti,ab,kw]
16. simulation environment [ti,ab,kw]
17. Interactive [ti,ab,kw]
18. reality system [ti,ab,kw]
19. 1 OR 2 OR 3 OR 4 OR 5 OR 6
20. 7 OR 8 OR 9 OR 10 OR 11 OR 12 OR 13 OR 14 OR 15 OR 16 OR 17 OR 18
21. 19 AND 20

Appendix 4. SinoMed (Chinese)

1. Breast Cancer
2. Virtual Reality
3. 1 AND 2

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	2
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	4-5
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	5-6
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	6-7
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	6
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	6 and Appendix A
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	8
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	8
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	8
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	8
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	8
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	Table 2
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	9
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	8
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	9
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	9
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	9
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	No
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	9
Certainty	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	9

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
assessment			
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	9 and Figure 1
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	Figure 1
Study characteristics	17	Cite each included study and present its characteristics.	Table 2
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	Table 1 and Figure 2
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	19-20 and Figure 3 4 5
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	19-20
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	19-20
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	19-20
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	19-20
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	19
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	Table 1 and Figure 2
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	21-23
	23b	Discuss any limitations of the evidence included in the review.	26
	23c	Discuss any limitations of the review processes used.	26
	23d	Discuss implications of the results for practice, policy, and future research.	24-25
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	3
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	Not registered
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	Not registered
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	27
Competing interests	26	Declare any competing interests of review authors.	27
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	In the manuscript and appendix

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
			document

From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71

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The Efficacy of Virtual Reality-Based Interventions for Breast Cancer Patients Symptom and Rehabilitation Management: A Systematic Review and Meta-Analysis

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Primary Subject Heading:	Oncology
Secondary Subject Heading:	Complementary medicine, Rehabilitation medicine
Keywords:	Breast tumours < ONCOLOGY, Information technology < BIOTECHNOLOGY & BIOINFORMATICS, REHABILITATION MEDICINE

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The Efficacy of Virtual Reality-Based Interventions for Breast Cancer Patients Symptom and Rehabilitation Management: A Systematic Review and Meta-Analysis

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Word count: 4,039 (excluding title page, abstract, references, figures, and tables)

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Abstract

Objectives To determine the effectiveness of VR-based intervention on the symptoms and rehabilitation management in patients with breast cancer.

Design Systematic review and meta-analysis.

Study selection We included all eligible randomized controlled trials (RCTs) and quasi-experimental studies (published in English and Chinese).

Participants Breast cancer patients (≥18 years) undergoing cancer treatment.

Interventions Any intervention administered to improve the symptoms and rehabilitation of breast cancer patients. The control group was given conventional care.

Outcomes All outcomes were as follows: pain, fatigue, anxiety, depressive symptoms, cognitive function, and range of motion of upper limb in breast cancer patients.

Data sources We searched PubMed, Embase, CENTRAL, and SinoMed, four electronic databases, covering the database establishment period to January 2022.

Review methods Two reviewers independently extracted content and data consistent with the prespecified framework and assessed risk bias. Random-effects meta-analysis was used to pool data across trials. Meta-analysis was performed using Review Manager V.5.4.

Results: A total of 8 studies met the eligibility criteria and were included in this study. The combined effect size showed that VR was positive for improving patients' anxiety(SMD=-2.07, 95%CI= [-3.81 to -0.34], I^2= 95%) and abduction of upper limbs(MD=15.54, 95%CI= [12.79 to 18.29], I^2= 0%), but fatigue(SMD=-0.92, 95%CI= [-4.47 to 2.62], I^2= 99%) was not. Qualitative analysis showed VR improved patients' depressive symptoms, pain and cognitive function.

Conclusions: Virtual reality technology has a good effect on breast cancer patients' symptoms and rehabilitation management, but the quality of evidence is low, and the sample size is small. To date, there are few intervention studies, therefore, giving precise recommendation or conclusion is difficult. We have a favorable view of this, and more clinical studies are needed in the future to improve the credibility of the results.

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Keywords virtual reality; breast cancer; meta-analysis

Strengths and limitations of this study

1. Unlike previous systematic reviews that focused on many types of cancer patients, this is the first review that aims at investigating the effectiveness of virtual reality in breast cancer patients to evaluate the relative safety and effectiveness of this new technology.
2. Due to the diversity of research design, we used the random effect model for analysis.
3. Faced with the COVID-19 pandemic period, the intervention based on virtual reality provides patients to undergo rehabilitation effectively at home and reduce the risk of infection.
4. High heterogeneity in some outcome measures due to different measurement tools, low quality of included studies, and small sample size limit the further promotion of the results, resulting in a risk of bias.
5. The quality of the included literature is low, the long-term effect on patients is still unclear, there is a lack of literature support with a higher level of evidence.

Systematic review protocol and registration None.

Patient and public involvement Not applicable.

Ethics Statement Not applicable.

1. Introduction:

Breast cancer is the most commonly occurring cancer in women[1]. There were over 2 million new cases and nearly 630 thousand deaths globally in 2018[2]. Moreover, according to the prediction, new breast cancer patients will increase by more than 46% by 2040[3]. In recent years, there has been a remarkable improvement in breast cancer-related survival rate and life quality[4], because of the accessibility to various treatment options and rehabilitation options, for instance, early breast cancer screening, chemotherapy radiotherapy, surgical therapy, and comprehensive treatment

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rehabilitation management[5-9]. However, the long-term breast cancer disease progression and the treatment have some adverse effects, which can cause some physical and psychological negative consequences[10]. These possible long-term adverse effects include somatic symptoms such as limited joint range of motion, pain, and psychological distress such as fatigue, anxiety, and depressive symptoms[11-13]. Simultaneously, these symptoms and negative emotions may further cause cognitive dysfunctions for the patient, leading to poor predictive outcomes and affecting all aspects of their life[14,15].

Some adjuvant treatments are used for rehabilitation management of breast cancer patients to improve symptoms, side effects, and the quality of life[16]. This process is often carried out through traditional physical therapy and occupational therapy to improve the patients' symptoms and perform rehabilitation management[17]. However, these traditional methods are time-consuming, cumbersome, and cost a lot, and the results often depend on the medical staff's actual capabilities[18]. Moreover, traditional rehabilitation therapy is primarily mechanical repetitive training, which lacks interest and motivation[19]. Patients often lose interest in this process, which affects the training effect of patients. So the limitations of traditional rehabilitation training prompted the emergence of new and effective methods.

With the rapid development of computer science, more and more modern technologies are applied to managing symptoms and rehabilitation of breast cancer patients, and it is continuously developing[20].

Virtual reality (VR) is mainly used as a brand new rehabilitation therapy for breast cancer patients. The concept of VR was first proposed in the 1960s and gradually rose at the end of the 20th century, and it is an artificial intelligence technology based on multidisciplinary cooperation[21].

It uses the computer as the core architecture, calculates and simulates a three-dimensional virtual environment, and combines with other somatosensory devices (such as somatosensory handle, sensing helmet, 3D glasses) to make up the gap between the virtual environment and reality[22].

It allows interactive feedback between the user and the virtual environment, providing

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the user with visual, auditory, tactile, and other signal stimuli to produce their feelings. The features of virtual reality are summarized into '3i'. They are immersive, interactive, and imaginative [23]. Compared with the traditional " viewing from the outside - in silico operation, "VR emphasizes human beings' dominant role in virtual environments. With the high-speed development of virtual reality technology, its connotation is also further broadened. Based on Virtual Reality (VR), the concepts of Augmented Reality (AR), Mixed Reality (MR) were extended[24]. Although virtual reality technology is more diversified, it is mainly applied to breast cancer patients in immersive experiences using headgear and computer games.

In recent years, VR has received increasing attention as a symptom and rehabilitation training approach to address pain reduction, fatigue, anxiety, depression, and cognitive dysfunction in different groups. It has been applied for physical rehabilitation management of some diseases, such as stroke, Parkinson's, rectal cancer, and prostatic cancer[25-28]. Moreover, it has indeed received good results, and the quality of life of these patients has been dramatically improved. Considering that breast cancer patients have similar physical and mental burdens, it is feasible to attempt virtual reality into them. As for breast cancer patients, virtual reality technology was gradually introduced into American scholars' clinical intervention research in the early 21st century[28]. However, there are few intervention studies for breast cancer patients. Many of them are still in clinical trials, and there is no systematic review to summarize and analyze them. There is no specific intervention strategy. Existing systematic reviews primarily focused on the inclusion and analysis of mixed samples of cancer patients[29].

Therefore, we need to focus on specific populations of patients to be selected in research. Thus, doubts could be raised as to whether VR treatment methods for breast cancer can improve these symptoms; and whether the rehabilitation of patients is effective is worth studying. For this reason, we aimed to perform a meta-analysis as a scientific method to investigate the effects of this uncertain treatment using statistical methods to examine whether VR training is practical for breast cancer patients' symptoms and rehabilitation management. According to the current guidelines for breast cancer patients, we mainly focus on those who receive breast surgery or receive radiotherapy

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or chemotherapy on TNM 1-4 stage[30]. Based on the data analysis findings, we can make an objective evaluation and analysis of virtual reality technology in the rehabilitation and symptom management of breast cancer patients, hoping to provide a reference for the Application of patients and the clinical decision-making of medical staff.

2. Methods:

2.1 Search strategy

We checked up to three English databases (Pubmed, Embase, and CENTRAL), and one Chinese database (SinoMed) from the establishment of databases to November 2020, with the following Mesh terms and text words: ("Breast Neoplasms" OR "Breast Cancer Lymphedema" OR "breast tumor*" OR "breast cancer") AND ("Virtual Reality" OR "Virtual Reality Exposure Therapy" OR "simulation environment"). Appendix A shows the exact searching strategy. Finally, based on the above results, we performed a snowballing approach to search, screen, and examine papers, classifying them as potentially eligible studies.

2.2 Eligibility criteria

2.2.1 Types of studies:

We included all eligible randomized controlled trials, and quasi-experimental studies (published in English and Chinese), at least one of which was an ongoing virtual reality training program for inclusion in the review.

2.2.2 Types of participants:

Our inclusion population was required to meet the following criteria: i. breast cancer diagnosed by pathological analysis (TNM stage I–IV); ii. age ≥ 18 years old; iii. after surgery ongoing chemotherapy or radiotherapy. We also developed exclusion criteria for the participants: i. brain primary (patients may have adverse reactions to VR); ii.

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combined with other severe diseases (such as severe heart disease, hypertension.); iii. previous psychiatric illness or cognitive impairment; iv. primary limb movement disorders; v. visual and auditory impairment.

2.2.3 Types of interventions:

The trial group participated in the organized and systematic intervention project based on virtual reality technology for breast cancer patients. The primary intervention goal was to reduce the psychological disorder, dysfunction, and pain of patients.

The intervention experiment based on virtual reality needs to meet the following core conditions:

1. Using computer-generated human-computer interactive intervention means in the virtual environment, such as VR glasses and headwear equipment;
2. Patients immerse themselves in the virtual environment constructed by electronic equipment;
3. Patients can perceive, feel, and interact in a manner that is similar to a physical place;
4. They can achieve effects by combining stimulation over multiple sensory channels such as vision, sound, touch, and perception to produce a corresponding stimulus.

And the potential comparison groups included any format of rehabilitation intervention, like exercise and usual care.

2.2.4 Types of outcomes:

Due to the different focus of the original research, the leading evaluation indicators may be various. To show the improvement of outcomes for patients by VR as comprehensively as possible and provide a reference for subsequent studies, we selected as many outcomes as possible under objective and fair evaluation conditions. The outcomes included: the patient's anxiety, fatigue, depressive symptoms, pain, cognitive function, and abduction of upper limbs. The measurement tools for outcomes are shown in the characteristics of the included studies.

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2.3 Data collection and analysis

2.3.1 Selection of studies

Two review authors independently screened all search results (title, abstract) in the database to identify studies that could be included in this paper. After initial screening, the review authors performed a full-text evaluation to determine the literature for inclusion. When disagreements occurred, they discussed the disparities and resolved them. If consensus was not reached after discussion between the two, it was decided by the third author independent arbitration. If necessary, we contacted study authors for specific information.

2.3.2 Data extraction and management

Two review authors independently extracted data onto a predesigned data collection form, including the trial setting, inclusion and exclusion criteria, study population, interventions, outcome measures, and final results of the papers. Both review authors, before data extraction, were given detailed instructions and participated in training sessions. Any disagreements or discrepancies regarding data extraction were resolved through discussion, and a third reviewer was consulted in case of any disagreement.

2.2.3 Assessment of risk of bias in included studies

We performed the risk of bias (ROB) of RCTs using Cochrane Collaboration ROB-2[31]. The overall assessment was recorded as high, low, and some concerns. Synthesis of ROB plots was done using online software Robvis (visualization tool)[32]. We used JBI critical appraisal checklist for quasi-experimental studies evaluated and included the before-after study[33].

2.3.4 Data analysis

All analyses were performed using RevMan V.5.4 (Cochrane Collaboration,

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http://ims.cochrane.org/revman). Heterogeneity was assessed using Cochran's Q test and quantified with the I^2-test. A random-effects model was applied to calculate the pooled results if $I^2 \geq 50\%$; otherwise, a fixed-effects model was used. Forest plot graphics were generated to present the pooled effect. The mean difference (MD) and standard MD (SMD) with corresponding 95% CI were used to calculate the effect size. SMD was calculated where the same outcome was reported but using different measurement tools. All tests are two-sided, and P-value<0.05 was considered to be statistically significant.

2.3.5 Dealing with missing data

There was no missing data for any of the included articles.

3. Results:

3.1 Search process

After searching in 4 databases, we retrieved 3,058 articles. For these documents, we conducted a duplicate check, preliminary screening, and full-text reading for screening. Finally, eight papers met the criteria of this meta-analysis. 6 were randomized controlled trials, and 2 were before-after studies. Figure 1 shows the detailed search process and results.

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3.2 Critical appraisal of quality

Figure 2 shows the quality assessment of the included studies: 3 RCTs[34-36] judged by ROB-2 have a high risk of bias, 1[37] have a low risk of bias, while the remaining 2[38,39] studies showed the uncertain risk bias. The following were the primary sources of bias: deviations from intended intervention, differences in outcome measurement, and unclear randomization process. In summary, the quality of the included RCTs was considered as having a high risk of bias. 2[40,41] before-after studies judged by the JBI tool show the central bias exists in: comparison receiving similar care and multiple measurements of the outcome (Table 1).

Table1. Results of critical appraisal for quasi-experimental studies

Questions (potential bias and threat)	House et al. [40]	Chen et al. [41]
1. Is it clear in the study what is the 'cause' and what is the 'effect'?	Yes	Yes
2. Were the participants included in any comparisons similar?	Yes	Yes
3. Were the participants included in any comparisons receiving similar treatment/care, other than the exposure or intervention of interest?	No	Unclear
4. Was there a control group?	Yes	Yes
5. Were there multiple measurements of the outcome both pre and post the intervention/exposure?	Yes	No

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6. Was follow-up complete and if not, were differences between groups in terms of their follow-up adequately described and analyzed?	Yes	Yes
7. Were the outcomes of participants included in any comparisons measured in the same way?	Yes	Yes
8. Were outcomes measured in a reliable way?	Yes	Yes
9. Was appropriate statistical analysis used?	Yes	Yes

3.3 Study Characteristics

The essential characteristics of the included studies are shown in Table 2. The quality of the eight studies met our inclusion criteria. Three studies were from China [35,36,41], and the others were from the United States[40], Jordan[38], Italy[34], Turkey[37], and Australia[39]. Two of them were before -after controlled trials, and the other six were randomized controlled trials. The research was conducted in specialized wards and cancer centers of hospitals. The number of participants ranged from 6 to 120. All participants were breast cancer patients who met the inclusion criteria. VR intervention included both immersive and non-immersive formats, and the duration of the intervention ranged from 20 minutes to 3 months. The studies assessed VR-based interventions' impact on health-related outcomes, including anxiety, depressive symptoms, fatigue, pain, cognitive function, and abduction of upper limbs.

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Table 2. Characteristics of the included studies

Study (Country/ year)	Design	Target population	Disease Stage /Age	Frequency (days/week)	Duration/ session (min)	Intervention time (weeks)	VR Method	Study Outcome	Study Results
House et al. [40] (USA/2016)	Before-after study	Post-surgical breast cancer women(n=6) with pain in the upper arm.	Unknown Age:57.8±20.4	2	20-50	8	The BrightArm Duo Rehabilitation System consists of a low-friction robotic rehabilitation table. A computerized forearm supports a display, a laptop computer for the therapist station, a remote clinical server, and a library of custom integrative rehabilitation games. It tracks arm position and grasping strength while patients play three-dimensional (3D) custom integrative rehabilitation games. Training difficulty increased progressively in game complexity, table tilt, and session length	BDI-II-depression symptom-decreased; BVMT-R/HVLT-R/TMT/NAB- cognitive function-improved; NRS-pain-decreased; Mechanical Goniometers-ROM-improved	PS PS PS PS
Chen et al. [41] (China/2019)	Before-after study	Breast cancer women (n=80) after surgery were hospitalized for chemotherapy.	II-III Age range:40-60	5	20-30	8	Patients with its motion-sensing controller and computer to build virtual cognitive rehabilitation training system, memory, executive ability, the training of the information processing speed and concentration of four modules, with the help of virtual scene and the virtual objects, through a variety of tasks such as puzzles, tai chi, a virtual maze game, improve patients' cognitive function, and real-time information feedback and rehabilitation information support.	MoCA -cognitive function-improved	PS
Mohammad et al. [38] (Jordan/2018)	RCT	Female patients with breast cancer(n=80) at a specialized cancer center	II-III Age:51.99±10.34	When giving morphine	15	When giving morphine	The CG (n=40) did not receive VR treatment. The VG(n=40) chose from two scenarios on a CD-ROM, which included deep-sea diving "Ocean Rift" or sitting on the beach with the "Happy Place" track. Then, the patients wore a head-mounted display with headphones. The PI remained near the participants during the VR session. The VR exposure session was ended at the peak time of painkiller efficacy.	VAS-pain-decreased; SAI-anxiety-decreased.	BS BS
Chirico et al. [34] (Italy/2019)	RCT	Female breast cancer patients(n=94) were receiving	I-III Age: VG:55.18±5.7 MTG:55.7±5.26	During each chemotherapy	20	During each chemotherapy	The CG(n=34) received routine care, and the MTG(n=30) received music therapy based on CG. The VG(n=30): The VR equipment consisted of a headset and an ahead movement tracking system. Participants explore the island through the relaxing landscape created	SAI-anxiety-decreased; SV-POMS(T/D/F)-depression symptom/fatigue-decreased	BS BS

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		chemotherapy.	CG:56.2±6.79				on the Second Life® platform through forests, animal observation, mountain climbing, and swimming.		
Feyzioğlu et al. [37] (Turkey/2020)	RCT	Breast cancer women(n=36) had undergone surgery.	Unknown; Age: VG:50.84±8.53; CG:51.00±7.06	2	45	6	The VG(n=19) received tissue massage, passive mobilization, and Xbox360Kinect™ video game treatment. The patient performed activities based on VR boxing, darts, and other games; the CG(n=17) received standard upper limb physical therapy treatment, including scar tissue massage and passive mobilization.	VAS-pain-decreased; Digital Goniometer-ROM-improved.	PS PS
Jimenez et al. [39] (Australia/2018)	RCT	Female breast cancer patients(n=37) received radiotherapy.	Unknown; Age range VG:35-74 CG: 35-74	Unknown	60	18	The VG(n=19) used the VERT system (Version 2.9) to provide basic technology, anatomical knowledge, and radiation dose information about RT for patients. The CG(n=18) received regular health education on radiotherapy knowledge.	STAI-S-anxiety-decreased.	PS
Jin a et al. [36] (China/2018)	RCT	Female breast cancer patients(n=120) undergoing surgery.	I-III Age range 20-50	6	30	12	The VG(n=60) used a professional rehabilitation platform; through the virtual and VR somatosensory equipment (helmet, joystick, data gloves), patients can choose their rehabilitation games for training. The CG(n=60) received traditional rehabilitation training.	SF-36-pain/fatigue-improved	BS
Jin b et al. [35] (China/2018)	RCT	Female breast cancer patients (n=76) received surgery.	II-III; Age range 33-69	14	15-30	12	The VG(n=38) selected appropriate training methods for rehabilitation according to the patients' muscle strength characteristics in different postoperative rehabilitation periods. The system includes sensors (HTC), upper limb rehabilitation devices, and a game platform (Unity 3D). The CG (n=38) received routine rehabilitation training under the guidance of nurses.	Mechanical Goniometers-ROM-improved	BS

Abbreviations: BDI-II: Beck Depression Inventory, Second Edition; BVMT-R: Brief Visuospatial Memory Test, Revised; CG: Control Group; HVLT-R: Hopkins Verbal Learning Test, Revised; MoCA: Montreal Cognitive Assessment; MTG: Music Therapy Group; NAB: Neuropsychological Assessment Battery; NRS: Numeric Pain Rating Scale; RCT: Randomized Controlled Trial; ROM: Range of Motion; SAI: State Anxiety Inventory; SF-36: 36-Item Short Form Survey; STAI-S: State-Trait Anxiety Inventory-State; SV-POMS(T/D/F): Short Version of Profile of Mood States in tense, depressive, and fatigue); TMT: Trail Making Test; VG: Virtual Reality Group; VAS: Visual Analog Scale.

BS: The comparison between the two groups was statistically significant.

PS: Outcome improved post-intervention, but intergroup results were not statistically significant, or there was no control group.

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3.4 Outcome analysis

Due to the limitation of research type and significant heterogeneity between groups, we only selected anxiety, fatigue, and abduction for meta-analysis. As for the remaining outcomes, we chose the form of qualitative analysis. We detailed the research form, time, equipment, and content of each study in Table 2. Although we could not conduct a meta-analysis, we also analyzed and expounded on their strengths and limitations.

3.4.1 Anxiety meta-analysis

Three studies[34,38,39] evaluated the impact of VR on breast cancer patients' anxiety. Compared with the control group, the results showed that VR-based intervention had a significant effect (SMD=-2.07, 95%CI= [-3.81 to -0.34], P=0.02, Figure 3). Heterogeneity was considerably high (I^2= 95%).

3.4.2 Abduction meta-analysis

Two studies[36,37] evaluated the impact of VR on breast cancer patients' abduction. Compared with the control group, the results showed that VR-based intervention had a significant effect (MD=15.54, 95%CI= [12.79 to 18.29], P<0.00001, Figure 4). Heterogeneity was considerably low (I^2= 0%).

3.4.3 Fatigue meta-analysis

Two studies[34,35] evaluated the impact of VR on breast cancer patients' fatigue. Compared with the control group, the results showed that VR-based intervention had no significant effect (SMD=-0.92, 95%CI= [-4.47 to 2.62], P=0.61, Figure 5).

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Heterogeneity was considerably high (I^2= 99%).

3.4.4 Depressive symptoms analysis

In two studies, participants were distracted by training in a recovery game in a virtual environment, and both before-after study[40] and randomized controlled trial[34] showed improvement in depressive symptoms. At the end of the intervention, the depression symptoms scores (MD) of patients in the before-after study decreased by 8.3 compared to baseline data($P<0.05$), and there were differences between the VR and control groups after the intervention in the randomized controlled trial($P<0.05$). The results were statistically significant.

3.4.5 Pain analysis

Three studies focused on patients' pain symptoms after the VR intervention, one before-after study[40], and two RCTs[37,38] showed a reduction in patients' pain scores. At the end of the intervention, patients' pain scores (MD) in the before-after study decreased by 1.5 compared to baseline($P<0.05$). There were differences in pain after intervention between the VR and control groups in two randomized controlled trials($P<0.05$). The results were statistically significant.

3.4.6 Cognitive function analysis

Two before-after studies[40,41] examined the effects of virtual reality on patients' cognitive function, showing significant improvements in executive function, language function, memory, and visual function. The results were statistically significant.

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4. Discussion

This systematic review and meta-analysis evaluated the application effect of intervention based-on virtual reality in breast cancer patients. Consistent with previous research[42], the results showed that compared with traditional breast cancer care, patients' anxiety, depressive symptoms, pain, cognitive function, and abduction of the upper limbs were improved with a statistical significance. However, the pooled effect showed that VR-based intervention had no effect on fatigue.

Our results showed that virtual reality had a significant effect on anxiety among patients with breast cancer. With the transformation from the traditional medical system to a biopsychosocial medical network, much more attention has been paid to the role of anxiety in the etiology and prognosis of breast cancer[43]. The Prevalence of anxiety among breast cancer patients was 41.9%[44]. The previous study assessed the effectiveness of VR for cancer patients, and their results showed VR had no significant effect on reducing anxiety in patients[29]. The difference in results might be that we focused on breast cancer patients, who were more uniform and had minor variations. Our results indicated that the heterogeneity of the included studies was significant. Due to the limited number of articles, we were unable to conduct sensitivity analysis to find the source of heterogeneity. According to the study design, we found that heterogeneity might exist in the specific treatment types of patients undergoing VR intervention, different pathological stages of cancer, small sample size, inconsistent equipment used, and different or unclear frequency and duration of intervention. Like anxiety, breast cancer patients usually get mental problems and physical symptoms. 97% of breast cancer patients report experiencing depression symptoms and pain. Our review also

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included the effect of VR on depressive symptoms. In both studies analyzed, the VR treatment had a reducing effect on depressive symptoms. Because of the different types of research and lack of data, we could not show their combined effect sizes. We believe the possible mechanism behind VR's effect on the above symptoms is the distraction mechanism. According to Buhle et al. [45], the mechanism of distraction is different from placebo. Their MRI analysis showed distraction effectively suppressed pain processing in the brain, while placebo had no significant effect on pain processing. We hypothesize VR reduces pain through distraction, thereby improving patients' psychiatric symptoms during treatment. As mentioned before, our study was limited in many aspects, such as the type of study and sample, so we should treat this result with caution.

Our results showed that virtual reality had a significant effect on the abduction of upper limbs among patients with breast cancer. Bleeding, effusion, necrosis of skin flap, and upper limb edema often occur after radical breast cancer surgery, which will cause upper limb dysfunction such as limited shoulder joint activity and limb weakness, and significantly reduce the quality of life of patients[46].VR as a rehabilitation tool has been very mature. Application in improving the range of motion of patients' upper limbs, especially in stroke patients, has been well verified[47]. Our results suggested that VR as a rehabilitation tool could improve some aspects of patients' upper limb movement. Still, we need more extensive sample size trials to confirm its long-term effects. VR may not be the only influential factor; it may need other auxiliary tools to best use its effects.

Cancer-related fatigue has been documented as one of the most distressing symptoms reported by breast cancer survivors. It affects functioning and impacts quality of life. Possible causal factors include physical conditions, affective and cognitive states, proinflammatory cytokines, and metabolic factors[48]. Our study found that virtual reality

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had no effect on fatigue among patients with breast cancer, which was contradictory to the findings of Zeng[29], who reported that VR-based intervention was related to a statistically significant decline in fatigue scores among patients with breast cancer. The possible reason for the discrepancy may be related to the differences in participants' characteristics. In addition, this may be related to the small sample size of literature included in the study, resulting in a significant bias. More strategies need to be explored to help breast cancer patients simultaneously manage the level of fatigue.

16% to 75% of breast cancer patients receiving chemotherapy have some degree of cognitive impairment[49]. Cognitive impairment seriously affects patients' quality of life, even more than the recurrence and metastasis of breast cancer itself. Two before-after studies examined the cognitive improvement of VR in breast cancer patients with improved vision, memory, executive function, and language function after the intervention. Due to technical reasons, we failed to conduct a combined effect size analysis, but from the data analysis of the before-after study, VR is beneficial to the cognitive function of patients. The primary characteristic of VR is immersion, which enhances the subject's heart rate and activates brain circuit functions. In particular, it acts on the midline θ of the brain's forehead, including the frontal lobe and the left temporal area, bringing a pleasant emotional experience to subjects, activating the attention of related brain networks, and improving motor integration and spatial orientation functions[50].

To sum up, in this context VR uses immersive experience to intervene and achieve positive effects for breast cancer patients. The devices which were used are: wearable helmets, 3D glasses, somatosensory operating devices, smart displays, and computers. Subjects can interact cognitively or physically in a computer-generated virtual

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environment in real-time to achieve their goals. An immersive tool changes patients' focus of attention, provides a positive and pleasant experience, and improves progress. Because of the interactivity of VR, patients enhance the effectiveness of an intervention in the real-time interaction of VR. This may be related to activating the midbrain marginal dopaminergic pathway and VR technology stimulation in the interaction between patients and VR[51]. The brain's reward mechanism can improve the attractiveness of participating in the intervention, enhance breast cancer patients' interest, and make patients more like this form. VR has the characteristics of real-time in virtual environment. Not only can real-time monitor the behavior and physiological response of patients, background data operation and guardians observe the transmission of stimulation in the virtual environment and comprehensively control the duration and intensity of rehabilitation[52]. The above utility mechanism improves the positive experience, attention, interaction, and participation of patients to a certain extent. It carries out rehabilitation in a real-time dynamic monitoring environment to improve patients' compliance, promote the rehabilitation of patients with breast cancer, and improve their prognosis.

Direction for future research and practice

The discussion shows VR for breast cancer symptom management and rehabilitation intervention methods are worth learning and promoting. With the rapid development of 5G communication technology, artificial intelligence, and Internet of Things technology, VR methods are bound to be more widely used and will receive technological innovation. Around the world, some countries have begun to formulate relevant laws and policies to give VR technical and economic support and policy inclination to accelerate industrial integration and put the new VR technology into

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clinical practice in medical and health care as soon as possible[53].

According to our research results, we state the following directions for further research. The Use of VR Security Question. When using a VR device, subjects are immersed in a virtual environment and often ignore the potential dangers in the actual scene. There is a risk of accidental injury and falling. Therefore, they should be monitored during use. At the same time, professionals should evaluate and modify their environment to ensure patient safety. Simultaneously, the articles retrieved in our systematic review rarely involve safety evaluation indicators, so we did not use them as outcome indicators. In the follow-up research, the safety evaluation system of VR application in patients can be constructed and applied. Be alert to the potential adverse effects of VR. The highly immersive nature of VR stimulates sensory nerves, which may lead to adverse reactions such as blurred vision, dizziness, nausea, and vomiting. Although the existing short-term intervention studies have not clearly defined it as an adverse event, long-term effect monitoring is still needed to avoid possible health problems and addiction caused by long-term use. The hardware and ecological construction are not yet mature. The Application of VR to breast cancer is still in its infancy. Most VR products on the market are not specifically developed for breast cancer patients. The clinical integration is low, ecological resource construction is not mature, and patients' targeted demands cannot be met. It can be low selectivity. It is suggested that a multidisciplinary team should be formed to develop VR hardware and ecological communities for breast cancer patients, build a more extensive resource library, enrich the functions of VR, and give patients more choices to maximize the benefits. During the COVID-19 pandemic, the social activities of the population were more alienated. This has demonstrated the importance of remote treatment systems when in-person

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patient-doctor contact is not possible. Because of the convenience and mobility of breast cancer intervention based on VR, patients can also be treated at home, reducing the risk of infection when going out and alleviating the feeling of social isolation. Telemedicine technology based on VR may become a trend to improve the utilization efficiency of medical resources.

Limitations

This study has some limitations that need to be addressed, and which is why we need to interpret the results with caution. First of all, the types of studies included were relatively diverse, so we could not analyze the combined effect size in some outcomes. Although the qualitative analysis was carried out and the results were positive, we should treat it with caution because there was no quantitative analysis result. Secondly, we clarified and discussed the staging of breast cancer patients. Most patients were in stage II-III, and the staging of patients in some of the included studies was unclear. The specific form of VR therapy that patients received also varies, so we cannot recommend its use for specific clinical populations, but based on our findings, we believe VR has significant benefits for patients with mental symptoms. VR interventions included in the study are different in terms of intervention time, frequency, and specific intervention form, so there is heterogeneity. The small sample size and number of participants in our study may lead to considerable differences between groups. And for this reason, we were unable to check for publication bias in this study. In the future, we suggest researchers conduct multi-center randomized controlled trials to seek higher-quality evidence to prove the effect of VR on breast cancer patients.

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5. Conclusion

The systematic review showed that VR-based interventions improved anxiety, depressive symptoms, pain, cognitive function, and abduction of upper limbs in breast cancer patients, but not fatigue. Due to this paper's limitations and extensive sources, we were unable to carry out specific recommendations and conclusions. In the future, large-scale pilot studies on symptom management and rehabilitation in breast cancer patients should be carried out to explore its applicability and feasibility in breast cancer patients.

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Figure legend/caption: Figure1 Flow chart for study selection according to PRISMA Declaration 2020. PRISMA, Preferred Reporting Items for Systematic Reviews and

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Meta-Analyses; Figure2 Quality assessment of the included studies; Figure3 Effect of VR-based interventions on anxiety; Figure4 Effect of VR-based interventions on the abduction of upper limbs; Figure5 Effect of VR-based interventions on fatigue.

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53 Valle-Cruz D, Gez EAR, Sandoval-Almaz R, Criado JI. A Review of Artificial Intelligence in Government and its Potential from a Public Policy Perspective. *Proceedings of the 20th Annual International Conference on Digital Government Research* 2019.

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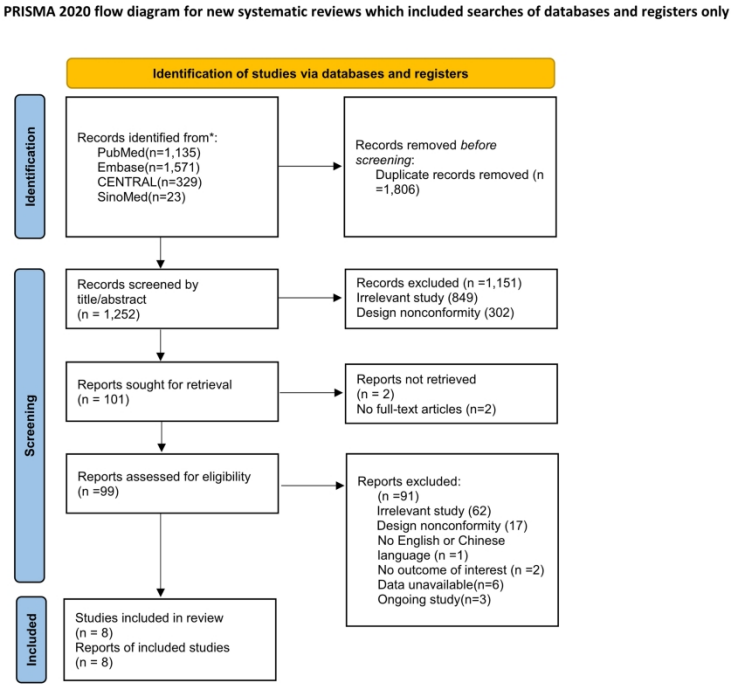


Figure1 Flow chart for study selection according to PRISMA Declaration 2020. PRISMA, Preferred Reporting Items for Systematic Reviews and Meta-Analyses

210x296mm (500 x 500 DPI)

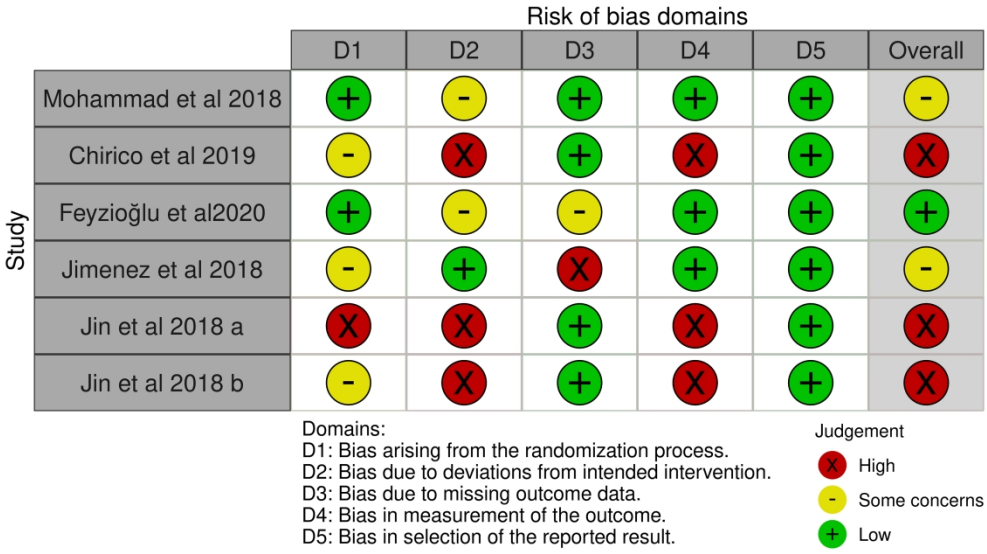


Figure2 Quality assessment of the included studies

447x283mm (197 x 197 DPI)

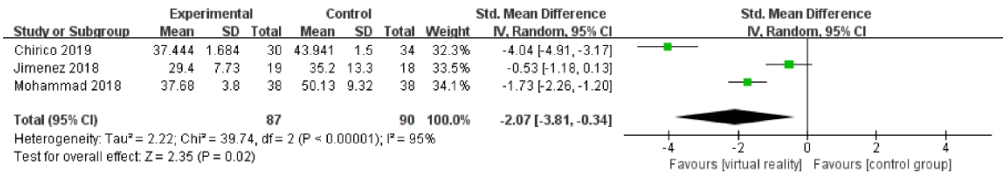


Figure3 Effect of VR-based interventions on anxiety

810x143mm (157 x 157 DPI)

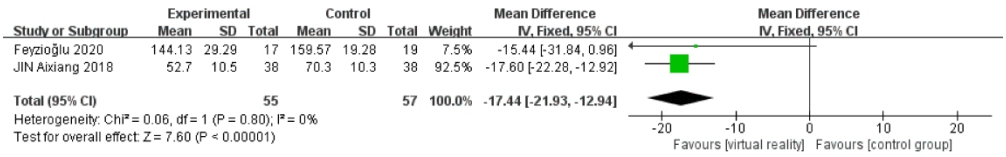


Figure4 Effect of VR-based interventions on the abduction of upper limbs

820x129mm (157 x 157 DPI)

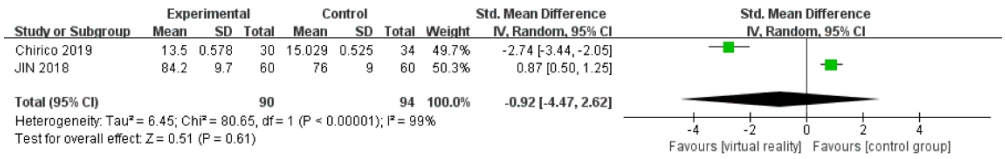


Figure5 Effect of VR-based interventions on fatigue

810x129mm (157 x 157 DPI)

Search Strategy

Appendix 1. PubMed (English)

1. Breast Neoplasms [MeSH Terms]
2. Breast Cancer Lymphedema [MeSH Terms]
3. breast tumor* [Title/Abstract]
4. breast cancer [Title/Abstract]
5. breast carcinoma* [Title/Abstract]
6. mammary gland carcinoma* [Title/Abstract]
7. Virtual Reality [MeSH Terms]
8. Virtual Reality Exposure Therapy [MeSH Terms]
9. User-Computer Interface [MeSH Terms]
10. virtual rehabilitation system [Title/Abstract]
11. virtual reality head-mounted display [Title/Abstract]
12. VR [Title/Abstract]
13. game [Title/Abstract]
14. gaming [Title/Abstract]
15. simulation system [Title/Abstract]
16. simulation environment [Title/Abstract]
17. 1 OR 2 OR 3 OR 4 OR 5 OR 6
18. 7 OR 8 OR 9 OR 10 OR 11 OR 12 OR 13 OR 14 OR 15 OR 16
19. 17 AND 18

Appendix 2. Embase (English)

1. 'breast cancer'/exp
2. 'breast tumor'/exp
3. 'breast cancer-related lymphedema'/exp
4. 'breast neoplasm*' [ab,ti]
5. 'breast carcinoma*' [ab,ti]
6. 'mammary gland carcinoma*' [ab,ti]
7. 'virtual reality'/exp
8. 'virtual reality exposure therapy'/exp
9. 'computer interface'/exp
10. 'virtual reality head mounted display'/exp
11. 'virtual rehabilitation system'/exp
12. 'vr' [ab,ti]
13. 'user-computer interface' [ab,ti]
14. 'game' [ab,ti]
15. 'gaming' [ab,ti]
16. 'simulation system' [ab,ti]
17. 'simulation environment' [ab,ti]
18. 'interactive' [ab,ti]
19. 'reality system' [ab,ti]
20. 1 OR 2 OR 3 OR 4 OR 5 OR 6

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21. 7 OR 8 OR 9 OR 10 OR 11 OR 12 OR 13 OR 14 OR 15 OR 16 OR 17 OR 18 OR 19
22. 21 AND 22

Appendix 3. CENTRAL (English)

1. MeSH descriptor [[Breast Neoplasms] explode all trees
2. MeSH descriptor [[Breast Cancer Lymphedema] explode all trees
3. breast tumor* [ti,ab,kw]
4. breast cancer [ti,ab,kw]
5. breast carcinoma* [ti,ab,kw]
6. mammary gland carcinoma* [ti,ab,kw]
7. MeSH descriptor [Virtual Reality] explode all trees
8. MeSH descriptor [Virtual Reality Exposure Therapy] explode all trees
9. MeSH descriptor [User-Computer Interface] explode all trees
10. virtual rehabilitation system [ti,ab,kw]
11. virtual reality head-mounted display [ti,ab,kw]
12. VR [ti,ab,kw]
13. game [ti,ab,kw]
14. gaming [ti,ab,kw]
15. simulation system [ti,ab,kw]
16. simulation environment [ti,ab,kw]
17. Interactive [ti,ab,kw]
18. reality system [ti,ab,kw]
19. 1 OR 2 OR 3 OR 4 OR 5 OR 6
20. 7 OR 8 OR 9 OR 10 OR 11 OR 12 OR 13 OR 14 OR 15 OR 16 OR 17 OR 18
21. 19 AND 20

Appendix 4. SinoMed (Chinese)

1. Breast Cancer
2. Virtual Reality
3. 1 AND 2

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	2
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	4-5
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	5-6
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	6-7
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	6
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	6 and Appendix A
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	8
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	8
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	8
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	8
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	8
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	Table 2
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	9
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	8
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	9
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	9
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	9
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	No
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	9
Certainty	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	9

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
assessment			
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	9 and Figure 1
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	Figure 1
Study characteristics	17	Cite each included study and present its characteristics.	Table 2
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	Table 1 and Figure 2
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	19-20 and Figure 3 4 5
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	19-20
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	19-20
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	19-20
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	19-20
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	19
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	Table 1 and Figure 2
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	21-23
	23b	Discuss any limitations of the evidence included in the review.	26
	23c	Discuss any limitations of the review processes used.	26
	23d	Discuss implications of the results for practice, policy, and future research.	24-25
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	3
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	Not registered
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	Not registered
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	27
Competing interests	26	Declare any competing interests of review authors.	27
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	In the manuscript and appendix

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
			document

From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71

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